



Biometric signature authentication using machine learning techniques: Current trends, challenges and opportunities

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Abstract

Biometric systems are playing a key role in the multitude of applications and placed at the center of debate in the scientific research community. Among the numerous biometric systems, handwritten signature verification has got keen interest over the last three decades. Handwritten signature verification is the behavioral bio-metric system that discriminates the genuine signature from the pre-stored known signatures. It has been researched in the number of application areas like banking, financial and business transactions, cheque processing, access control and e-business etc. In this article, we surveyed the techniques of offline and online signature verification systems according to the taxonomy of classification model. A detailed background of signature verification system along with the available datasets are presented comprehensively. At the end, we presented the most notable challenges that guide the readers towards the current trends and future directions of the domain.

Keywords Biometric · Online signature verification system · Offline signature verification system · Classification models

1 Introduction

With the rapid development in information technology, biometric systems have been evolved as a revolutionary security systems in pattern recognition and E-systems [222]. Biometric systems are the security systems that tends to identify or verify the individual on the the basis of their physiological or behavioral traits [131]. Basically the word *biometric* is the amalgamation of two Greek words: bio and metric. *Bio* means life and the *metric* means to measure [49]. The biometric systems were primarily presented in 1970s for the law enforcement agencies to identify criminals through their fingerprints [25, 205].

Although biometric systems deals with individuals unique characteristics for the recognition and authentication, thus it can be categorized into two broad characteristics i.e.,

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physiological and behavioral characteristics [28, 217]. Physiological characteristics are static in nature that contain unique features of human body. These characteristics include physiological modalities like fingerprints, thumb impression, palm print, veins, iris recognition, face recognition and DNA etc. [178]. On the other hand, behavioral characteristics are dynamic in nature that may change with the passage of time, mood, age and other factors. The modalities involve in behavioral characteristics are gait [124, 185, 187], signature, handwriting [176, 177], voice, keystroke etc. [178]. Figure 1 illustrates the physiological and behavioral characteristics of biometric. It can be observed that the verification based on behavioral characteristics are lucrative but difficult than physiological characteristics.

Biometric information system is either based on verification system or identification system. The verification system aims to verify the individual's distinctiveness based on his or her professed individuality. On the contrary, identification system creates the individual's identity (between individuals registered in expert system) without the person appealing their specification. In simple words, verification system is one-to-one (1:1) search while identification system is one-to-many (1: N) search system [22, 115, 165]. These biometric systems are playing a vital role in different areas to enhance the security levels and act as suitable key segment in swarm of approaches like bank check processing, writer identification [177], online banking, face detection [110], medical detection [119], attendance, official document and security purposes etc. [164, 232].

Despite of several biometric systems, one of the most demanding and leveraging behavioral biometric system is the signature verification. The term signature is derived from Latin source *signer* means to sign. A signature is any handwritten specimen in a someone's own writing that defines the individual's first name, last name, surname, nick name or combination of them used for identification [38, 142]. The signature verification system is process of discriminating genuine and forged signer from the pre-stored known source samples [147]. It is one of the most preferable biometric verification system due to the dominating properties like user-friendliness, socially and legally unobjectionable by the society, non-invasive in addition to large development in numerous applications.

Handwritten signature verification has been comprehensively considered in this modern era. A signature verification method can be categorized into two main classes: online and offline signature verification [13, 104]. Online signature verification system measures the sequential data by capturing the signature image using tablets, PDA's, magnetic pad, smart phones. It is dynamic in nature that operates on dynamic features like direction of writing, positions of pen tips, velocity, order of strokes, speed and pressure etc to [91, 127, 141].

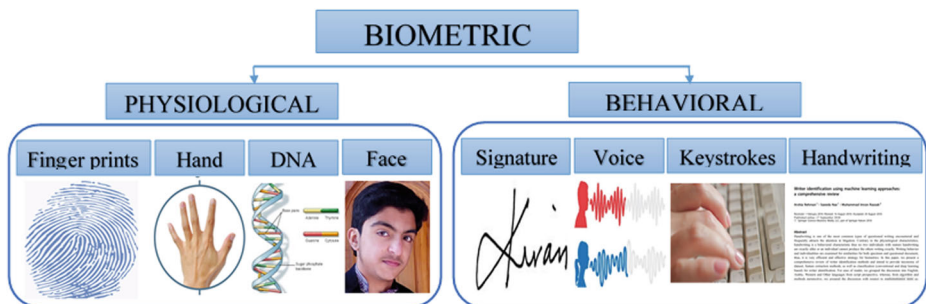


Fig. 1 Physiological and behavioral characteristics of biometric

Alternatively, offline signature verification system uses an optical detector to take signature on the page, image or document is scanned. This method is static in nature that contains static information like slant, bounding box, length and height of signature, baseline, pressure and size [155]. Offline signature verification is relatively more difficult as compare to online signature verification due to absence of dynamic parameters detail [131]. Therefore online signature verification system provide better accuracy and often higher recognition rates than off-line verification system. The workflow of offline and online signature verification systems are depicted in Fig. 2.

The basic intention of signature verification system is to differentiate the classes of signature based on variation [48]. There are ultimately two distinct classes of signatures such as genuine and forged signatures. *Genuine signature* is the signature that belongs to an original person. *Forged signature* includes the copied or imitated signature of person created by illegal person. One of the key factor of signature verification system is to identify the forged signature in order to reduce the crimes and hacking information. Thus the mechanisms of signature verification system outcomes the type of the signature forgery [80, 171]. Forgeries of signature can be further categorized into three classes like random forgery, simple forgery, skilled forgery.

In *random or zero effort forgery* [78, 218], person does not have information about name of the original person or signature shape. *Simple forgery/ casual forgery* is a non-professional forgery in which the name of original signer is known but the knowledge of signature shape is unknown [78, 218]. However *skilled or simulated forgery* is a qualified imitation of the genuine signature. In this type of forgery, forger has knowledge about the name of the original signer and its shape therefore it is harder to detect [63, 218]. Apart from these forgery classes, other classes had been introduced like unskilled forgery [63], targeted forgery [48, 70], tracing forgery, cut-and-paste forgery, electronic forgery, freehand signature forgery [37]. These forgeries have been rarely focused however a detail information can be found from the work of Deore et al. [37] and Nguyen et al. [135]

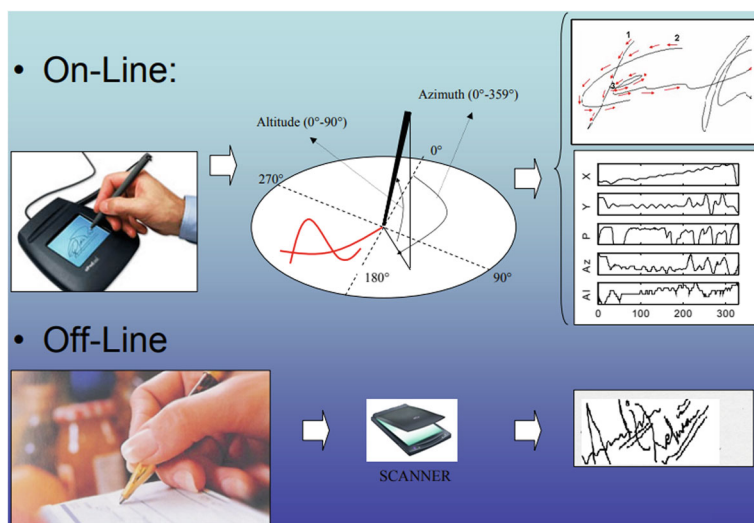


Fig. 2 Workflow of offline and online signature verification

The objective of this paper is to provide an in-depth review of signature authentication systems to the readers. In contrast to several reviews on this subject, our review paper provides additional information which is lacking in other review articles, such as:

- Provides detail background about the signature verification.
- Critical analysis evaluation on the literature review of signature verification.
- Provide an in-depth information about the datasets for the researchers in the domain of signature.
- Review on the signature verification system according to the taxonomy of classification models.
- Tabulate datasets and literature review statistics in the comprehensive manner.
- Contains current market challenges along with future prospects for signature verification system

Rest of the paper is organized as: Section 2 elaborate the background of signature verification system. Section 3 discuss the several databases corpus in online and offline domain of signature verification. Traditional phases of signature verification are explained in Section 4. Then Section 5 will presents the literature review of online and offline signature verification system according to the taxonomy of classification models. Section 6 will give the challenges offering in the domain along with the future perspectives. Finally the conclusion will be drawn in Section 7.

2 Background

The study of signature verification is quite an old concept tracing back to the 439 AD, when signatures were used for authentication of documents in Roman Empire. In 1677, England passed an act for the validation of signatures to prevent frauds [46]. The Roman Empire was one of the primitive government that presented laws for information regarding dispute points by professionals. After two centuries ago, it was permitted in English-speaking in 1792. These laws remain in progress until the Common Law Procedure Act approved in 1854 at England. In earlier time handwriting was physically verifies and the method itself is tedious and cost effective. Later in the 20th century, as computer was invented, the analyst tried to develop applications to verify handwritten signature to track the restrictions of conventional approaches. Automated signature recognition is the way to deal with stable machines to analyze or justify the handwriting and signatures.

Signature verification is developing rapidly in the information technology since 1960 [46]. First automated signature recognition system was presented in 1965 by North America Aviation [57]. The first off-line and on-line signature verification system was presented by Nagel and Rosenfeld [121] and Liu and Herbst [65]. In the following of these studies, varieties of methods and algorithms have been proposed to tackle this problem [207]. In 1965, Mauceri first published the work in on-line signature identification. In 1966, Kozinets et al. [211] first time employed computer for the off-line verification system. In 1973, Nagel and Rosenfeld [121] published first offline signature verification algorithms in IEEE conference and transaction [122]. As well as Liu and Herbst [65] published online signature verification algorithm in 1977. In 1986, Ammar et al. suggested a Pseudo-dynamic feature extraction procedure to extract pressure info from gray scale images. In 1987, Sabourin and Plamondon expressed first study for proposed method. In 1993, the strategy was proved in the known case of Daubert vs. Merrell Dow Pharmaceuticals [211]. In 2004, the Hong Kong

university conducted the international competition for online signature verification for first time [229].

Despite of the fact that number of work has been done in the field of offline or online signature verification in the past few decades. Number of attempts have been presented since 1960's, but Plamondon and Lorette [161] presented the first survey of signature verification and writer identification in 1989. After this, number of review papers published in the literature listed in Table 1. It is concluded and cleared up from Table 1 that the available survey papers discussed the offline or online signature verification system, separately. It could not find a review or survey paper that could provide an in depth detail about the both verification systems. Another limitation is about the detailed information of the background history. These survey papers only discuss that signature verification system is an old concept since last three decades. However the question arises how much old concept it is. This survey paper provides the detail background history of signature verification system. The

Table 1 Summary of the sate of the art surveys on signature verification and identification

1989	Plamondon and Lorette [161] discussed the work of automatic signature verification as well as writer identification since 1989 in his survey paper comprehensively.
1992	Sabourin et al. [182] presented another survey to discuss the perspectives of identification based on signature images.
1994	Leclerc and Plamondon [100] explained the work of automatic signature verification form 1989 to 1993.
1997	A review of online signature verification including some earlier work on the offline case by Gupta and McCabe [56]. Sabourin [181] gave another survey that describes the advancement on offline signature verification.
2000	Advanced survey from 1980 to 2000 is discussed by Plamondon and Srihari [162]
2004	Dimauro et al. [39] further discussed advancements of signature verification in his survey paper. Weiping et al. [66] published his survey in 2004 for offline signature verification.
2008	Another detail survey was conducted in 2008 by Impedovo and Pirlo [71].
2009	The survey on Online handwritten signature verification by Venkatesha et al. [170]
2011	The state of the art from year 2000 to 2010 was published by Wilkin , and Yin [224]. S Pal et al. [147] presented a survey in 2011 for offline signature verification. Pal et al. [146] published his survey in order to discuss the verification of signatures other than Latin script. Al-Omari et al. [11] discussed offline signature verification systems in his survey paper.
2012	In 2012, survey on Offline signature recognition and verification was conducted by Singh and Sharma [202]. Another signature verification survey was published in 2012 by Sanmorino and Yazid [183].
2013	Offline Signature Verification survey conducted by Yadav et al. [227]
2014	In 2014, survey on off-line signature recognition and verification schemes was published by Vala and Joshi [213]. Ubul et al. [212] discussed the technologies of offline signature verification systems using Non-western Script and another survey on offline and online signature verification was published by Yogesh and Patil [230].
2015	Deore and Handore [37] presented a survey on offline signature recognition/ verification methods. Pawar [157] discussed approaches for signature verification. Bhumika and Kumar [98] also presented a survey on signature verification techniques. Pawar and Dhope [156] presented a survey on static signature verification and recognition using NN. Another survey in this year was presented by Mohammed et al. [118]. In this year, survey on the Online Signature verification and Recognition had been published by Shabir et al. [76], Aswathy [20], Yogesh [231].
2017	Hafemann et al. [61] further presented a survey in the domain of offline signature verification. Another Survey on Offline Handwritten Signature Verification by Malphedwar et al. [220]. Another contribution in this year was given by Tamlarasi and Kalyani [210] using signature verification algorithms.

motivation of this survey is to provide an in-depth review of signature authentication systems to the readers. In contrast to several reviews on this subject, our review paper provides additional information which is lacking in other review articles, such as Provide an in-depth information about the datasets for the researchers in the domain of signature, review on the signature verification system according to the taxonomy of classification models, tabulation of datasets and literature review statistics in the comprehensive manner, contains current market challenges along with future prospects for signature verification system verification.

3 Statistics of dataset

A dataset is a collection of logically related data of discrete items stored in a single database table or in a statistical data matrix. In any research domain, dataset availability is one of the central requirement for the development and evaluation [5]. In the domain of signature verification, we have significant dataset in different scripts explained in the forth coming subsections and tabulated in Table 2 for the ease of understanding.

3.1 UTSig

UTSig stands for "University of Tehran Persian Signature" is a publically available Persian dataset that comprises 8280 samples from 115 classes. Every class has 27 genuine, 3 Opposite-hand and 45 skilled forgeries from 6 forgers. The contributors are randomly

Table 2 Publically available handwritten signature Corpus's

Dataset names	Languages	Types	Users	Genuine	Forgery
UTSig [203]	Persian	Offline	115	27	45 skilled forgeries
FUM signature [117]	Persian	Offline	20	20	10
MCYT [143]	Spanish	Offline	75	15	15 skilled forged
MCYT-75 [50]	Spanish	Offline	75	15	15 skilled forgeries
MCYT-100 [143]	Spanish	Online	100	25	25
BIOSECURE-DS2 [67]		Online	104	20	30
GPDS signature [111]	English	Offline	160	24	30 forged
GPDS-960 [215]	English	Offline	960	24	30
Sigcomp2009 [27]	Offline	Offline	12 genuine ,31 forgers	5 genuine signatures per authors	5 forgeries per authors
SigComp2011 [102]	Dutch and Chinese	Online/ Offline	10/10	235(Dutch) 240(Chinese)	340(Dutch) 123(Chinese)
SVC2004 [229]	English and Chinese signatures	Online	40	20	20

chosen from Iranian undergraduate and graduate students of University of Tehran and Sharif University of Technology.

3.2 Fum

A FUM is a Persian signature dataset that comprises of 20 classes. Each class contains 20 real and 10 forged samples.

3.3 MCYT

MCYT Bimodal Subcorpus offline signature dataset contains 2250 signatures acquired from 75 users using "WACOM INTUOS A6 USB". Each user have 15 genuine and 15 skilled forged samples.

3.4 MCYT-75

MCYT-75 is a sub-corpus of MCYT bimodal offline dataset that consist of 2250 signature samples collected from 75 signers. Each class comprises of 15 real and 15 forgeries by 3 forgers. Entirely the database contains 1125 genuine and 1125 forged samples. Forgers had real images and they copied their shape and natural dynamics.

3.5 MCYT-100

MCYT-100 is an online signature dataset that include 25 genuine and 25 skilled forgeries of 100 signers.

3.6 BIOSECURE-DS2

The BIOSECURE-DS2 contains dynamic signatures collected in two sessions from 104 contributors (30 genuine and 20 forgery signatures per contributor).

3.7 GPDS signature

The GPDS signature database is offline dataset that contains 2106 signatures composed by 39 persons. The dataset has 160 classes and each contains 24 authentic and 30 forgeries imitated by 10 forgers. Participants used black/blue ink and a white papers having 2 different box sizes.

3.8 GPDS-960

GPDS-960 is an offline signature dataset. Signatures were taken on paper with 24 various box sizes. The dataset contains 24 genuine and 30 forged signatures collected from 960 individuals. Therefore, the dataset contains total 23049 genuine and 28800 forgeries which were performed by 1920 people other than 960 above specified. Each forger filled a form of 15 boxes and each form have 5 samples of randomly selected genuine signatures. Each forger had 5 genuine samples from 5 specimens which was imitated by 3 times.

3.9 Sigcomp2009

The ICDAR2009 Signature verification competition was held on 10th International conference on document analysis and recognition (ICDAR'2009). The signatures were acquired in both online and offline mode. The offline training dataset contains 1920 samples from 12 real authors and 31 forgers. As a result of error, while documentation the online, training set includes only 1905 documents. The evaluation set was captured at the Netherlands Forensic Institute, which consists of genuine samples from 100 newly chosen writers and forged signatures from 33 authors. Dataset have overall 1953 signatures for both the online and the offline dataset.

3.10 SigComp2011

SigComp2011 consist of both online and offline signatures. It include Dutch and Chinese signatures in which the training set include 235 genuine, 340 forged signature for chinese, 123 forged signatures for Dutch from 10 specific authors. Similarly testing set contains 116 reference, 120 questioned and 367 forgeries from 10 specific chinese writers while 648 references, 648 questioned and 638 forgeries from 54 distinct Dutch writers.

3.11 SVC 2004

SVC2004 dataset is in the online mode, containing both English and Chinese signatures, captured using WACOM Intuos tablet. Dataset consist of two tasks using 2 database. The first database contains 20 genuine, 20 skilled forgeries samples for 40 writers. 5 other volunteers contributed in making four random forgeries for each. The second database consists of similar signature data collected from 60 users. First task is suitable for signature verification using PDA and second task is for digitizing tablets.

4 Stages of signature verification system

The signature verification system has several traditional phases but firstly it is decided that approach is online signature verification or offline signature verification. The traditional phases of signature verification comprises of data acquisition, pre-processing, feature extraction and classification. These main phases of signature verification system are graphically illustrated in Fig. 3.

4.1 Data acquisition

Data acquisition means to capture the signature image [39]. Data acquisition is the primarily step for any verification method. Due to data acquisition, the handwritten signature verification system can be categorized as either: offline (static) system or online (dynamic) system. For offline system, signatures were collected using ball pens, gel and sketch pens having black and blue ink on A4 paper and respectively each individual provide different samples of his/ signatures. These signatures are then converted into gray scale with suitable resolution. Data acquisition is performed using cameras and scanners after the writing process has been completed. These scanned images are than pre-processed and stored into database in JPG or JPEG or PNG format. Each set in the database contains genuine and forged signatures used to train and test the model.

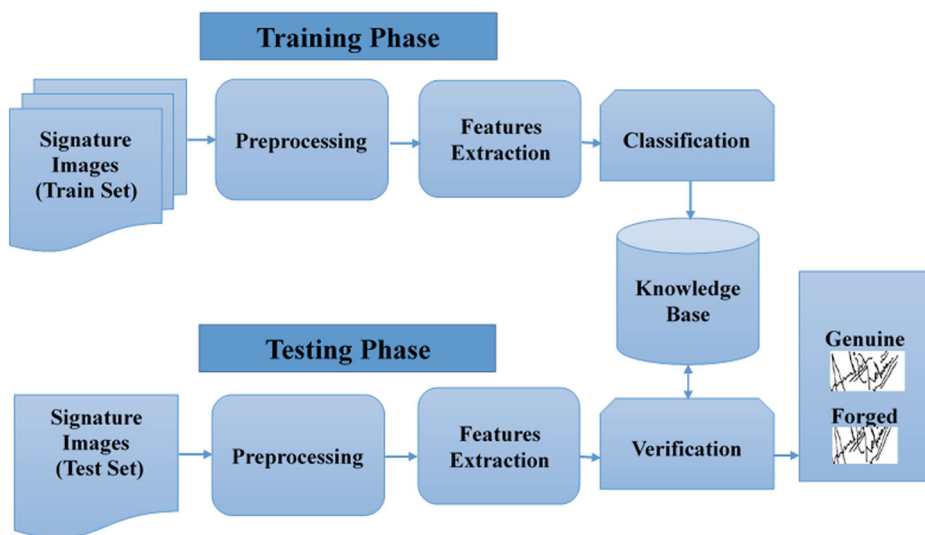


Fig. 3 Steps of signature verification system

Contrarily to online verification system, pen tablets, graphical tablets and touch mobiles are used for signature acquisition. The most commonly used data acquiring devices are the digitizing tablets. These particular devices create electronic signals, characteristics of the signature trace and overall the writing process. Ordinary generated signals are position, pressure, acceleration, velocity and force signals. Once the signatures are acquired it cannot be considered for further processing as it is. It needs some pre-processing that will be explain in the coming section.

4.2 Pre-processing

Pre-processing means to simplify certain transactions without losing significant processing [22]. Pre-processing is elementary step to enhance the input data or to sort raw data samples in a standard form that is appropriate for feature extraction phase.

In case of offline signature verification systems, images contains noisy pixels, complex background, blurriness and low contrast. Thus different pre-processing techniques were deployed to generate an enhance image that will be appropriate for further tasks. Typical pre-processing step consists of signature extraction, background elimination, signature alignment, grey scale conversion, noise removal by using filters, signature size normalization, binarization, width normalization, skeletonization, cropping, and thinning etc [165, 221].

Conversely, in case of online signature verification system, images may contain several types of variations and jerks that need to be removed in pre-processing phase. The pre-processing procedure can be accomplish through filtering, noise reduction, and smoothing algorithms.

Pre-processing of offline signature verification is relatively more complex and time consuming as compare to online signature verification system. The main reason might be signature segmentation due to the variations in handwriting, overlapping of characters, resemblance of different letters, loops, etc. in offline signature verification systems. In other

words, we can say that online signature verification systems are more efficient and accurate due to pre-processing strategies.

4.3 Features extraction

Feature extraction is the method of extracting the representative characteristics of signatures that are used for the discrimination of different signature classes [11]. The optimal selection of a dominant set of feature is the key for an effective system. Signature verification system has been measured from many different perspectives, yielding several alternatives for feature extraction. Broadly speaking, the feature extraction methods can be classified as offline/static or online/dynamic. Offline/static features extract the static information from the signature images like signature height, width, density, slant, baseline etc. Whereas online/dynamic features attempts to extract the dynamic information from the signature execution process such as position, pressure, acceleration, and velocity etc. Depending on the online or offline methods, there are three broad categories of features: statistical features, structural features, model based features or automatic features [176]. Statistical features are the mathematical or statistical measurements for classification of relevant information for reducing the gap among different classes.

Another broad categorization of statistical features are between Global and Local features [73]. The global features refers to a whole signature image or representative of the entire image. Common global features are displacement, area, compactness, acceleration, total signing duration, no. of pen ups/downs, total no. of components, signature global orientation, coefficients obtained using mathematical transforms, time duration of positive or negative position etc. In local feature extraction, signature image is divided into number of units or sections and the features are extracted from particular section of image. Local features can be further divided into two parameters like Component-oriented features and Pixel-oriented features [73]. In component oriented, the parameters are extracted at component level like height or width ratio of the stroke, strokes position, orientation of strokes, geometric features, orientation based features, slant, contour based and moment based features etc. On the other hand, the pixel oriented features are obtained at each level of pixel. Examples of pixel oriented features are grid based information, pixel consistency, gray-level intensity, texture features, shape based features, gray level intensity based features and shadow code based features etc.

The structural features are the local structure of the signatures like Local Binary Patterns (LBP), pixel density within signature grids, Scale-Invariant Feature Transform (SIFT) [3] and Speeded Up Robust Features (SURF) etc. There has been an increased attention in recent years on methods that do not rely on hand-crafted features. The model based or automatic features are learned from raw data (pixels, in the case of images) using specific models like Convolutional Neural Network (CNN) [126, 130], Extend Learning Model (ELM) [184, 201], Recurrent Neural Network (RNN) [4, 128, 129] etc.

4.4 Classification

Classification is the process to evaluate the authenticity of the query signature. The purpose of classification is to matching the features of query signature with the pre-stored knowledge base features for the sake of verification using large number of instances of reference signature based in a train set. After the training, the query signature is classified as genuine or forged. Typical classification techniques in the domain of signature verification include

rule based approach, template matching, Support Vector Machine (SVM), Neural Network (NN) etc.

After having discussed the traditional steps of signature verification system, we present notable recent research contributions to this domain in the following section.

5 A review of signature verification systems

Signature verification is one of the most active research problem in pattern recognition. As discussed earlier, the techniques developed for verification of signatures are generally divided into two categories on the basis of data acquisition: offline signature verification and online signature verification. In the subsections, we give an insight review about offline signature verification and online signature verification using different approaches.

5.1 Offline signature verification

The number of approaches and techniques were applied by number of researchers for achieving high performance, best accuracy and efficiency in the area of offline signature verification. The techniques established for signatures verification are commonly divided into three types; template matching, statistical approaches and structural approaches. We discuss the different verification methods proposed under each of these classes in the following.

5.1.1 Template matching

Template matching is the process of template/figure comparison in which test signatures are matched with templates already stored in database. Dynamic Time Warping (DTW) is the most common algorithm used for this purpose. Some of the brief review of contribution of researchers deploying DTW are discussed here.

Parziale et al. [153] conducted experiment on MCYT-100 and BiosecureID-SONOF. They extracted 8 statistical features like pen position, pressure, velocity, the pressure derivative and the acceleration. Z-score normalization was deployed and Euclidean distance was used for the comparison of feature vectors. Finally the classification was performed using DTW with two types of normalization. An EER of 1.30% on random forgery and 3.09% on skilled forgery was achieved using MCYT-100 database. The reported EER on BiosecureID-SONOF dataset was 1.17% on random forgery and 1.45% on skilled forgery.

Kumar in [95] verified the signatures using CEDAR database of 2640 signatures samples. They deployed trace transform to extract the invariant features. A feature vector of six features were created using six circus functions and then normalization is performed. The similarity measures for same-writer and different-writer pairs were performed using 0.069 threshold value. They reported ERR, FAR and FRR of 24.4%, 24.58% and 25.83% respectively.

Shanker and Rajagopalan [190] presented a signature verification system based on DTW. Initially signatures samples were preprocessed and vertical projection features were extracted. They modified the DTW and performed classification on the database of 100 persons. An ERR of 2% and ERR 29% is achieved respectively.

Kennard et al. in [88] implemented a 2D geometric warp algorithm for signature verification. To align the strokes of an examined signature with reference, 2D geometric warp was computed using automatic morphing correspondence algorithm. They verified the signatures using the distance measures. The obtained EER on English dataset of blind forgeries

was 94%-96%, 87%-91% on simple forgeries and EER accuracy on Chinese dataset was 74%.

5.1.2 Statistical approaches

Statistical approaches are the techniques used for the statistical analysis of data obtained from the statistical features of signature images. Some researches used statistical approaches to consider the relation and difference within the information of two or more signatures. Some of the statistical features in the domain of signature verification are distance measures, Hidden Markov Model(HMM), Genetic Algorithm (GA), Multilayer Perceptron (MLP) [186] and Random forest etc. A major contributions of the researches using statistical approaches are enlightened in this section.

A. Distance measures

Among the several statistical approaches, one of the simple and effective approach is distance based classification. This approach is free of model training and thus having reduced complexity. Normally, the approach is applied between the query and reference signatures. Euclidean distance, city block distance, Chi-square distance, Manhattan distance, and Hamming distance etc. are the most commonly techniques in this domain.

Davda and Gonsai [36] verified the signatures using two distance measures. They performed image inversion, noise removal, thinning, skew detection and correction in pre-processing. Matlab function was used for image inversion. Hough and affine transform was used for skew detection and correction and Bwareaopen function was used for noise removal. Images were resized using thinning operations. They extracted the local features using grid level and block energy. The FRR and FAR of 2.25% and 2.25% using city block distance and 2.5% and 2.5% using square Euclidean distance.

Du et al. [41] proposed a multi-stage approach that is Locality Sensitive Hashing (LSH) using DS-I and DS-II signature datasets. The Shape context features extracted from the pre-processed contour images. Frequency of each signature calculated for the representation. They classified the signatures using hamming distance and reported 84.15% accuracy in top 10 with low computational time achieved. Al-juboori in [9] conducted experiment on 108 signature samples from volunteers. Global thresholding was used for the conversion of grey scale images and median filter was applied for noise removal. Signature edges were detected using canny edge detector. Seven Hu invariants moments of 5 samples were calculated and feature vector was created by finding standard deviation. Euclidian distance was used for the comparison of saved and new feature vector. Azmi et al. in [21] conducted experiment on MCYT database. They computed statistical features like FCC features, height, aspect ratio, diagonal distance of signatures and centers of mass of all foreground pixels. KNN and Euclidean distance was used for classification. They reported lowest FRR was 6.67%, lowest FAR was 12.44% and AER was 9.56%.

Malik et al. in [107] used SigComp09 and 4NSigComp10 datasets for experimentation. Images were scanned, binarized and normalized using image processing techniques. They extracted 9 local features and performed classification using Gaussian mixture model (GMM). For global features, signature images were spatially smoothed and divided into 64 small parts. K nearest neighbor was used for decision of global feature vector. They reported EER of 20% for global, 36% for local features using SigComp09 and 20% for global features, 56% for local features using 4NSigComp10 respectively. Garhawal in [52] worked on a database of 90 signatures. They split the dataset in such a way 54 genuine signatures were used for training, 18 genuine and 18 forged samples were used for testing. They extracted

the SURF descriptors from images and calculated the intra class Euclidean distances, the intra-class thresholds, image distances between the test signature and pre stored template signatures. In order to measure the forgery, the intra-class threshold (like maximum, average, minimum and range) was compared to the inter class threshold. The specificity and sensitivity of maximum distance was 38.89% and 77.78%, average distance was 50% and 44.444% and the minimum distance was 38.889% and 44.444%. The specificity and the sensitivity of minimum/maximum intra-class distance using threshold value of 0.05 was 33.3%, 88.8% and 72.2%, 50%.

Neela in [133] introduced a method suitable for mobile devices which have touch interfaces. They performed time normalization and stroke concatenation in pre-processing. First they convert the time-series data of signature images into the sequence of cartesian vectors and then calculate the statistical features like x-y trajectories, speed, angles, and their derivatives. Manhattan distance method was used for matching the template signature with the query signature for verification process. Experimental results demonstrated the good performance of the proposed algorithm.

B. Hidden Markov Model

Just like other statistical approaches, hidden markov model (HMM) was widely used for signature verification. The following studies discussed the implementation of hidden markov models for signature verification. Coetzer [32] developed a system for the identification of handwritten signatures using the discrete Radon transform and HMM. Experiment was conducted on Dolfing's database having 924 signatures and an author generated database of 4800 signatures. The obtained ERR on skilled forgeries and simple forgeries was 18% and 4.5% using 924 signature database. The obtained EER on skilled forgeries was of 12.2% using author generated database of 4800 signatures. Justino and Yacoubi [79] extracted grapho-metric features like the density of pixels and the axial slant and then passed to HMM for classification. They performed 16-pixel horizontal segmentation with 25 cells. They calculated density of pixels by counting right afterwards pixels of cells and axial slant by skeleton in grid. Vector Quantization was used to convert vectors into sequence symbols. 20 signatures were used for learning of HMM, 10 for validation and 10 for verification.

Kaur and Choudhary [85] performed the signature verification on Punjabi database of 50 individuals. They performed thinning, noise reduction and image rotation in pre-processing. They extracted geometric features using Gabor filtering and statistical features using SURF descriptors. For training and testing of the signature verification system, 200 signatures were used. They performed the classification using HMM and showed the accuracy rate of 97%. Daramola and Ibiyemi [34] conducted experiment on 500 genuine signature images collected from 250 students of Covenant University Ota, Nigeria. Signature images were pre-processed using morphological operations. They extracted the features using DCT and calculated 16 DCT coefficients from the segmented signature blocks. Baum-Welch algorithm was used to maximize the observation feature vector of 16 DCT features. The successful signatures recognition rate of 99.2% achieved.

C. Neural Network

Neural networks are the set of algorithms, modeled loosely after the human brain, that are designed to recognize the signature patterns. This approach is widely used in signature verification systems because of the characteristics like power, ease of use, capabilities in learning and generalization. A lot of implementation had been done in signature verification using this approach explain below. McCabe et al. [112] employed neural network for the

verification of handwritten signatures. System was trained using 5 genuine and 100 zero-effort forgeries. Static and dynamic features were extracted. The proposed method reported the ERR of 3.3%.

Shikha and Shailja [199] proposed the method for offline signature recognition and verification using MLP and SOM neural network. They used 60 samples in such a way 20 genuine, 20 forged and 20 random forged signatures for experimentation. Signature image was normalized and skeletonized in pre-processing. They extracted statistical features like area, width, height, vertical and horizontal center of the signature and Eigen values. The Self Organizing Map (SOM) was chosen as learning algorithm and mapping was built by unsupervised training. MLP was used for supervised training. The performance of algorithm was calculated on different iterations (100, 102, 103, 104, 105). The minimum error rate of 18.5% obtained in iteration 103. Poap et al. [163] introduced the flexible neural network to assigned transformed objects as input in matrix form. Input curve was simplified by determining average points in each vertical row. For each coordinate arithmetic mean was calculated. All points were interpolated by Chebyshev method. 400 original signatures and 200 forgeries were used. 75% signatures were used for training set and 25% for verification set. The sigmoid function was used for activation with a threshold value of 0.3 and learning coefficient of 0.4. For verification, 30 randomly selected signatures were used by classical neural network and flexible neural network. Flexible approach reported correct classification at the level of about 85%. They achieved 17% increase in precision for original signature and 26% for fake signatures.

Sharma [196] proposed Linear Discriminant Analysis (LDA) and neural network for detection of forgeries. LDA was used to reduce the edges of the image. Neural network was used for training and testing of signature images. Input signature image was preprocessed and features were extracted from it. The features were mapped to the LDA. Neural network was used for the classification. The proposed algorithm results was FRR=0.0366 SSIM=0.8777 MSE=66.2507 PSNR=79.8582. Katiyar et al. in [82] acquired at least 10 specimen of each person signature. Scanning, cropping, binarization, compliment of binary image, noise reduction, clutter removal and region of interest were performed in pre-processing. They calculated local and global features. 3 signatures were used for training. The ANN had been used for classification. They achieved FAR of 3.2%, FRR of 2.2% and the resultant errors of 0.12%.

Kumar and Dhandapani [94] proposed experiment on CEDAR database. Signature images were scanned and covered by circles. Centroid of the signature image was computed. From the center of each circle, number of black pixels were calculated. Pixel distribution ratio of each circle was calculated. Extracted features from signature images were scale invariant and rotation invariant. Feed forward back propagation neural network was used for training, testing and validation of three sets of signature containing 24 genuine and 24 forged signatures. Average accuracy of training function was 97.89%.

Dongare and Ghongade [40] introduced artificial neural network for bank cheque signature verification system. Signature images of 10 individuals with forged signatures were scanned and preprocessed using different image processing techniques. Extracted features were stroke, moment invariants, GLCM, color dominant and histogram. The training and testing was performed using Artificial Neural Network. Algorithm and accuracy was not discussed in paper.

Padmajadevi and Aprameya [144] discussed simple method for verification of signature by calculating angle based features in signature samples. MCYT signature database was used and samples were preprocessed. For features extraction set of stocks had been obtained for given signature. The pixel tracking algorithm had been applied four times for each set a_i

of labelled pixels (0o, 40o, 60o, 80o, 100o, 120o, 140o, 160o, and 180o) in the four different orientations. SW, W, S, SE, NW, E, N and NE. The area, center of gravity, eccentricity and skewness along with angle based features were extracted. Feed forward back propagation neural network was employed for analysis. The recognition rate were controlled by threshold parameter. Accuracy of 97.61% was achieved. Kedia et al. [86] introduced a method of an automated offline signature recognition approach using new techniques. Signature images were acquired from individuals. Gabor filter was used for preprocessing of signature images and then applied linear discriminant analysis on the images. For signature matching, artificial neural networks used which were trained by the back-propagation learning algorithm. The average accuracy of a given method was 99.5% along with this the FRR rate was up to 73% and computational time of that technique was 0.87s.

Kumar et al. [97] conducted experiment on GPDS signature database. They performed binary conversion, thinning, resizing and bounding box of the signature in preprocessing. A feature vector of 24 statistical features like maximum horizontal and vertical histogram, center of mass, normalized area, aspect Ratio, tri surface, six fold surface and transition features etc. were computed. These features were fed to the Neural Network for classification. Classification rate of 82.66% was achieved. Karouni et al. in [81] worked on the database of 100 signatures. Color inversion, filtering and binarization was performed in preprocessing stage. Low pass FIR filter was applied to remove high frequencies. Simple shape based geometric features were extracted that were area, center of gravity, eccentricity, kurtosis and skewness. ANN was used to verify and classify the signatures. The error function MSC (mean-squared error) was used to minimize average errors. Finally the obtained FAR and FRR rate on threshold 90% were 1.6% and 3%.

Mehra and Gangwar [113] presented Surf features technique and neural-fuzzy techniques for signature recognition. For this purpose the scanned images were preprocessed. Simple geometric features were extracted that were area, center of gravity, eccentricity, kurtosis and skewness. SURF features and Neural Fuzzy techniques were implemented in Matlab using ANFIS tool. GUI window was created. Neural network learns the correlation of patterns and recognized the handwritten signatures based on the surf features. They concluded that this technique was better than previous techniques and given an accuracy of 98%. Huang in [69] described the work on standard MCYT-75 database. The signatures were taken by WACOM Intuos (Inking pen). Database contains 15 genuine and 15 forgery signature gathered from 75 users. 180 signatures (8 genuine and 7 forgery) of 18 users were used for training and 360 signatures (8 genuine and 7 forgery) were used for testing phase. Firstly signature images were preprocessed. The geometrical features were extracted and passed to neural network as an input to verify the signatures. Simulation was performed on Matlab R2013. The FAR and FRR of proposed method was 10.62 and 10.91 and total obtained accuracy was 89.24%. Abikoye in [1] performed background elimination, width normalization and thinning in preprocessing. Global features, mask features and grid features were extracted and applied as input for training phase. 8 different images were given to neural network algorithm to generate output. Proposed recognition system showed 100% success and also presented poor performance due to poor quality and high similarity between signature samples.

Sthapak et al. in [206] worked on GPDS signature database and to train the system 19 genuine samples taken from 30 individuals and 19 forgeries signature. The signature images were scanned and pre-processed. Stentiford algorithm was used for thinning process. The global features, mask features and texture features were extracted to create feature vectors. Trained the neural network and performed pattern matching and classification. 100-105 iterations used to check error rate. Reported FAR was (15, 12.5, 12.5, 14.5, 17.5), FRR was

(12, 15, 10, 12, 15) and TER was (27.0, 27.5, 22.5, 26.5, 22.5). Anand [16] worked in Matlab to enhance the signature verification and recognition. Preprocessing were applied on the acquired signature images then global and statistical features were extracted. 8 genuine signatures were used for training and some skilled forgeries was used for testing. To enhance the processing performance of different neural network such as back propagation, self-organizing map etc were compared but Cascaded feed-forward back-propagation networks showed best results. Finally 85-100% efficiency was achieved on different data.

Choudhary and Gcoe [31] used the database of 675 signatures of 56 persons to train the neural network. Database contains 8 genuine, 4 forged signature used for training and 112 genuine, 112 forged signature samples for testing. In preprocessing, signature images were normalized at size of 50×50 pixels. For smoothing and restoring, median Filter was employed. Thresholding technique was used for background elimination. Vertical and horizontal (histogram) projections were used for border clearing. They extract the translation, scale invariant and rotation normalization features using Invariant central moment and Zernike moment. These features were then passed to neural network for classification. The recognition system provided the 98% success rate. Kaur [84] described work on NFI Sig-Comp2009 dataset. Zernike Moments was extracted up to the order 4 from the digitized image as global shape descriptors, a ranking method (also called a filter based feature selection method) was utilized to select a no redundant feature subset with maximum recognition power on 11 dimensional features. Finally these features given to a multi-layer perceptron (MLP) based neural network. Mutual Information method used for variable ranking and choosing the most useful features. 10 fold cross validation method was used .To test the performance .The best performance was shown by first 8 ranked features with an EER value of 13:42% and area under the curve Az equal to 0:84.

Jarad and Tedmori [77] presented verification using the Back-propagation neural network algorithm. The system was trained on Intel core2 due 2.5 GHz machine with 2GB memory. A set of 900 signatures were used to train the system. The speed and throughput were 6s and the average FAR, FRR, and ERR of proposed system was 0.10, 0.04 and 0.415. Dash et al. [35] proposed offline signature verification using Adaptive Resonance Theory type-2 (ART-2) implemented in C language. 12 signatures were used for training and testing the neural network. The mismatch threshold was 5%. Through parametric trainings the best $= 0.97$ and $m = 20$ were obtained. The accuracy is found to be 99.98%. Mehta et al. [114] performed binarization and noise removal in preprocessing. They extracted spatial features of sum graph and classified using HMM and ANN. The lowest rate of correct classification of 86% and the highest is 92% was achieved.

Hatkaret al. [64] conducted experiment on 1000 signatures from which 5 genuine and 5 forged samples of 100 individuals were used for training. 24 test signatures were used to train the Neural network. If the resultant neuron produced value close to +1, test signature was stated as genuine and if it generates value closest to -1 it was generated as forged. The Accuracy of the system was 86.25%. Odeh and Khalil [139] used GPDS300signature database. Initially signature images were scanned and pre-processed. The eccentricity, skewness, kurtosis and orientation features were extracted from the pre-processed images. They trained the Neural network using back-propagation algorithm and retrieved the accuracy rate of 64% and error rate of 36% on training set. Accuracy rate and error rate for test set was 78.8%, 21.2%. Furthermore an overall error rate of 21.2% was reported.

Lakshmi [99] performed resizing, gray scale conversion, noise removal, binarization, thinning, extraction of the black pixels from the grid, defining center of mass and signature rotation in pre-processing. They extracted statistical features and fed to the Neural network. For simulation MATLAB NN-tool was employed. They used 15 signatures for training the

Neural network and 5 signatures for testing. The average FRR and FAR of 8%, 12% with a mean square error of 0.00314 was achieved. Pansare and Bhatia [151] extracted geometric features from pre-processed signature images using horizontal splitting technique. The Neural network was trained using error back propagation training algorithm. The achieved correct classification rate was 85.7% and Correct Classification Rate (CCR) was 100%.

Akram et al. [6] presented a comparative study of signature recognition using statistical features. They used 4000 signature samples collected from 40 individuals in their experiment. The extracted statistical features from pre-processed features were fed to the different Neural networks. The signature recognition comprises of feed-forward-back propagation neural network, competitive and probabilistic neural network. The highest accuracy of 100% accuracy was achieved using probabilistic neural network. Sharma and Khanna [195] presented an offline signature verification using supervised and unsupervised neural networks. They extracted seven statistical features and passed to the neural networks. The database of 300 signatures was used to train and test the feed-forward and Recurrent Neural Network. In case of supervised neural networks, threshold of 50% was set. The obtained CCR was above 95%, FAR was 2-4% and FRR was 3-5%. However in case of unsupervised neural network the CCR of more than 50% FAR was 2 - 4% and FRR was 13-15% was reported.

Puri and Verma [15] deployed random transform for feature extraction. They used 100-150 signature samples for training the back propagation neural network. All the output images had been collected by comparing the new images with already stored images and approximately 98.03% accuracy had been obtained by the system. Mukherjee et al. in [120] designed a neural network using auto associative memory for the signature verification. The signature images was scanned, resized and used for training data. The images were read in Matlab and converted into the matrices of binary. Then weight was assigned to each image. Total weight was calculated by summation rule. They passed random image to the activation function that matched the image with the existing database. If the image was present, it display 'known' as well as the name of the person. Arduino device was used with the Matlab output by glow the led for the verification of signature. This method reduced the time and cost complexity.

Pansare in [150] used GPDS data set for experimentation. The feed forward neural network was trained using Error Back Propagation Algorithm (EBPTA). The number of neurons in first two layers were 24 and 1 in the last layer. After feeding the feature vector of test signature, if the output neuron generated the value closest to +1, test signature was declared as genuine. Furthermore, if it generated the value close to the -1, it was declared as forged. They performed two experiments to evaluate their study. In first experiment, when the neural network was trained with 570 genuine and 570 forged signatures, it recognized 570 signatures as genuine and 570 as forged signatures. As a result FAR and FRR of the system was 0%, therefore, the Correct Classification Rate (CCR) is 100%. In second experiment, when the neural network was trained with 150 signature out of which 120 was genuine and 30 was forged signatures, it classify 22 signatures as genuine and 128 signatures as forged. The FAR rate of system was 14.66% and FRR rate of system was 20%, hence the generalized Correct Classification Rate of 82.66% was retrieved.

Malekian et al. [106] scanned the signature images and performed pre-processing. In pre-processing, median filter was applied for noise removal, complement of the image was taken for binarization and dilation was performed for thinning. They extracted statistical features like aspect ratio, horizontal to diagonal length ratio, the slope of the line joining the center of gravity and geometric center, vertical and horizontal variance based on shape and dimensions. Authors developed their own database that contain 900 signatures collected from 45 people. 300 genuine and 300 forged signature were used for training purpose, the

remaining 300 signatures comprised of 150 genuine, 150 forged were utilized for testing among 150 genuine samples 144 signatures were classified as genuine and 6 signatures as forgeries. Out of the 150 forged signatures tested with the network, 8 signatures were classified as genuine and 142 as forgeries. Levenberg–Marquardt algorithm was used for learning the neural network classifier. FRR of the system was 4.0%, FAR was 5.3% and overall CCR of the proposed method was 99.17% for the training set and 90.7% for the testing. Also MATLAB with Intel core i5 processor, 2.27 GHz and 4 GB RAM running Windows 7 was used to calculate best, worst and average response time.

Armand et al. in [18] presented an off-line signature verification using structural features that were extracted from the signature's contour to train the network. Experiment was conducted on database of 2106 signatures (936 genuine, 1170 forgeries). The unique structural features like (centroid, length, surface area, skew) and modified direction feature were extracted. The accuracy was achieved using Resilient Backpropagation (RBP) neural network and a Radial Basis Function (RBF) network. They reported accuracy of 91.21% and 88.0%.

5.1.3 SVM

A Support Vector Machine (SVM) is a discriminating statistical classifier formally defined by a separating hyperplane. In other words, the labeled training data is classified with an optimal hyperplane algorithm. In two dimensional space this hyperplane is a line dividing a plane in two parts where in each class lay in either side. Like other statistical approaches, SVM was also the focus of attention for the researchers of the domain. Most notable work of SVM is presented in this section. Shekar et al. [198] extracted the features using Taylor Series Expansion (TSE). They used CEDAR and MUKOS dataset and performed classification using SVM. They retrieved the accuracy of 95.25% on CEDAR dataset and 98.93% on MUKOS dataset respectively.

Sharif et al. [191] conducted experiment on CEDAR, MCYT and GPDS synthetic databases. In pre-processing, otsu segmentation, normalization, binarization, morphological operations and median filter techniques were employed to retrieve the images. They extracted horizontal, vertical and global features. The best features were selected with the aid of Genetic algorithm. SVM was used for the verification of signatures. They reported FRR, FAR and AER of 4.67%, 4.67%, 4.67% using CEDAR dataset, 3.67, 6.67, 5.0% using MCYT dataset and 4.16%, 3.33%, 5.75% using GPDS synthetic dataset correspondingly. Okawa [140] performed binarization and contrast enhancement in preprocessing. They deployed BoVW model and VLAD encoding with KAZE features for feature extraction. Experiment was performed on CEDAR and MCYT-75 datasets with the reported Error rate of 1.0% and 6.4%.

Kiani et al. in [92] discussed offline signature verification using local radon transform and support vector machines. Binarization and margin removal was done using Otsu binarization algorithm and inverting images using Radon image Transform. Local windows were used to segment signature image. Line segment detection was calculated by radon transform that computes projection sum of the image intensity with respect to angle and line. Line segment existence validation was calculated by comparing peak value with threshold. Feature vector extraction calculated by creating a feature vector for line width of 6 pixels. Feature vector normalization was done by dividing the elements to maximum value. SVM was used for the classification of signature images. Persian signature dataset was used for verification that consists of 30 signatures per 20 classes. 96% accuracy with (FRR of 4%) and FAR of

17% was achieved. Stellenbosch dataset that contain 22 classes, 10 genuine signatures are used for training and 20 genuine signatures for testing. FRR of 19% achieved.

Kiani et al. in [92] deployed Otsu binarization algorithm for the binarization of images. Feature vector was generated by applying radon transform that compute the projection sum of the image intensity with respect to angle and line. Line segment existence validation was calculated by comparing peak value with threshold. Feature vector normalization was done by dividing the elements to maximum value. SVM was used for the classification of signature images. Persian signature dataset was used for verification that consists of 30 signatures per 20 classes. 96% accuracy with FRR of 4% and FAR of 17% was achieved. They also used Stellenbosch dataset that contain 22 classes, 10 genuine signatures. FRR of 19% was achieved. Patil and Hegadi in [154] proposed discrete daubechies (db4) wavelet transform for the classification of offline signature. Wavelet coefficient and detail coefficient were extracted for three directions that were horizontal, vertical and diagonal. Standard GPDS database were used. For training set 4 genuine and 4 forgery signature had been used, and all the remaining signature of persons were used for testing set. The signature images was pre-processed and features were extracted using discrete daubechies (db4) wavelet transform. Each signature represents 80 features. For fast optimization Sequential Minimum Optimization (SMO) technique in SVM classifiers had been used. For implementation Matlab software had been used. In case of linear kernel FAR and FRR were 13% and 10% respectively whereas in case of nonlinear kernel FAR and FRR were 15% and 12%.

Pal et al. in [147] worked on English and Bengali scripts. A database of 1800 (1100 English+700 Bengali) signature samples was used for training and 846 (454 English+392 Bengali) signature samples for testing. Foreground and background information was used for feature extraction. To compute 400-dimension foreground/background under-sampled bitmap features and 400-dimensional gradient feature first normalized the input images by resizing image into 200×800 and 126×126 pixels and also boundary box found for samples were founded. The normalized image of 400-dimensional under-sampled bitmap features was segmented into 100 and images of gradient features into 16 overlapping blocks. Also 400 modified chain-code direction features was obtained the chain code frequencies for all 4 directions were computed. Gray-scale local-orientation histogram of an image Component was used for gradient features. Robert's filter was applied to obtain the gradient image, arc tangent of gradient is quantized into 16 directions also a Gaussian filter applied to obtain $5 \times 5 \times 16 = 400$ gradient features. The Euclidian distance measure used to calculate the distance. Nearest Neighbor (NN) techniques was considered as classifiers for signature identification. The result showed that 99.41% best results was obtained using 400-dimension chain-code direction features and the SVM classifier also 200-dimension under-sampled bitmaps features also demonstrated good results.

Randhawa in [172] worked on database of 500 signatures consists of 300 signatures genuine (30 signatures each of the 10 signers) and 200 forged signatures (20 signatures each of the 10 users). The signatures were scanned using HP-scan jet 5400c at 300dpi. Dataset was pre-processed by normalizing, grayscale conversion, skeletonizing and binarizing the samples. In the proposed system 28 features were extracted by Hu Moments and zone features by using Image Processing Toolbox (IPT) of Matlab 6.0. 400 signatures were used for training and 100 signatures for the testing purpose. Signature database was utilized for training the SVM. Then the signature verification accuracy of the model has been evaluated in terms of FAR, FRR and FIR. Accordingly, SVM described in this paper successfully verifies the off-line signature with 9% accuracy.

Hu and Chen in [68] worked on publicly available GPDS-300 dataset and Chinese dataset. In purposed system three separate pseudo-dynamic features based on gray level:

local binary pattern (LBP), gray level co-occurrence matrix (GLCM) and histogram of oriented gradients (HOG) were used both in writer dependent and writer-independent mode. The writer dependent Support Vector Machine (SVMs) classifiers was used to performed classification. In SVM both the database divided into 3 groups, every group consist of 100 individuals. SVM was trained with the obtained feature vectors from reference signature and random forgeries for each signer. The performance was measured using the skilled forgeries of the GPDS signature dataset and Chinese signature dataset CSD. The obtained results using GPDS and Chinese signature dataset were 5.66% and 9.94% equal error rate. Kruthi and Shet in [93] proposed an approach based on SVM for signature verification. A total of 336 signatures were acquired from different persons using ball, gel and sketch pens in separate templates. The scanned images were pre-processed using image processing techniques like binarization, complementing, filtering, thinning and edge detection. Using these pre-processed samples, statistical features like centroid, center of gravity, calculation of number of loops, horizontal and vertical profile and normalized area were extracted and stored in a database. Then SVM was used to classify the signatures and the overall classification error rate of 7.16% was achieved.

Parodi [152] used GPDS300 Signature CORPUS database from each writer 24 genuine signatures was used for training and 30 forged signatures for testing. Graphometric features were extracted from a circular grid. In order to achieve the robustness, rotation was mapped into the DFT. A SVM based classifier was used for classification of samples. The achieved FRR, FAR, EER of proposed approach were 7.82% 0.49% 4.21%. Vargas et al. [216] introduced a new method for verification of signature based on gray level information using three different databases (MCYT-75, GPDS100, and GPDS750). Posterization procedure was used for background removal. After that using Matlab function DTW2 a single-level 2-D wavelet decomposition was done. Wavelet analysis was performed for evaluation and improvement of global effect of ink-type. Statistical texture features were extracted using GLCM. The proposed method was trained using SVM and reported the ERR of 13.50% in comparison with other methods.

Randhawa et al. [173] extracted concentric squares based geometric features, zone based slope and slope angle features. The feature set consist of 102 extracted features. Image Processing Toolbox (IPT) in MATLAB 7.0 was used for signature pre-processing. The LIB-SVM 3.0 software was used to train the SVM model for verification task. SVM was used with three different types of kernel function. Radial Basis Function (RBF) based SVM model showed the finest outcomes as compared to the linear and polynomial kernels. The proposed system achieved FAR of 1.25% and FRR of 1.66%. Nguyen and Blumenstein [134] introduced a new grid based feature extraction technique using 2D Gaussian filter. The GPDS-960 corpus database was used. Signature contours were computed in pre-processing. The learning and classification processes were performed using SVM. The proposed system obtained AER of 13.90% and the FAR of 0.02% for random forgeries.

Pushpalatha et al. [167] introduced offline signature verification with polar feature descriptor of signature. On a registered class signatures, PLS Regression was applied to achieve regression score. Then Log Likelihood of the class signatures was calculated using Hidden Markov Model. If the regression score and Log Likelihood distance deviation was below 5%, it justify the validity of the classification. Multiclass Support Vector Machine was used for verification. An accuracy of proposed method was 98% with FAR of 8%. Suggested method identified random forgery with an accuracy of 76% and skilled forgery with an accuracy of 71%. Pal et al. [148] combined global filtering and G-SURF techniques for

feature extraction. The method was established on GPDS database. The learning and classification processes were performed using SVM. The achieved AER of proposed method was 2.95% and FAR, FRR was 3.55% and 2.35%.

Hafemann et al. in [60] worked on GPDS-960 and Brazilian PUC-PR. The dataset was partitioning into two sets that were development set and exploitation set for feature learning and enrollment. The CNN worked as a feature extractor. Images were resized using bi-linear interpolation with OTSU's algorithm. Normalization of the images was done by padding the images with white background. Linear SVMs and SVMs with the RBF kernel was applied for the writer-dependent classification. Normalization obtained good classification results on GPDS-160 dataset with a performance from 14.64% of EER to 10.70%. They recorded 15.05% EER with 4 signatures in the GPDS dataset, and 9.83% EER with 5 signatures on the Brazilian dataset. In [59], MCYT-75, CEDAR, GPDS-160 and the Brazilian PUC-PR datasets were used for experimentation. They extracted handcrafted and CNN based features and fed to the linear SVM for classification. They evaluated the biometric attacks to detect the forgeries and measured two two countermeasures on the success rate of the attacks.

5.1.4 Structural approaches

In the structural approaches, symbolic data structure such as trees, graphs and strings are used to represent the patterns of the signature. In this type of system, the symbolic representation will be compared to prototypes stored in the database. Structural methods are used in finding an equivalence aligning between two images. It shows good performance on detection of genuine signatures and forgeries.

Pal et al. in [145] worked on Hindi signature database. 16 genuine signatures were considered for training and 8 genuine signatures and 30 skilled forgeries for testing purpose. A histogram-based threshold technique and mean filter was applied for binarization. Total 24 features were extracted. An interval-valued symbolic representation-based approach was used for verification. A similarity measure was proposed as the classifier in this paper. The highest verification accuracy of 91.83% and FAR, FRR and AER rate of 2.5%, 13.84% and 8.17% was reported. H. N. and D.S [107] proposed bi-interval valued symbolic representation of signature. Histogram based global threshold was applied for binarization. The relative orientation and distance geometric centroid were used as features. MCYT-75 signature data set was used. For training set 5, 7 and 9 genuine signature of each individual were used while remaining used for testing. Max and Mean algorithm used for verification performance using different centroid. If the acceptance count of test signature was greater than the threshold (predefined) then the test signature was considered to be genuine otherwise forger. The lower AER (AER = 17.33 for "MEAN" fusion and AER = 18.26 for "MAX" fusion) than the approaches which directly use either distance features or orientation features.

Abuhaiba [2] presented graph matching problem for signature verification. 75 genuine signatures were scanned as binary images using an hp ScanJet 3400C scanner. Signature images were preprocessed using image processing techniques. They achieved EER of 5.7% and 26.7% on random and skilled forgeries. Shah et al. [188] discussed the handwritten signature using pixels intensity levels. MCYT dataset was used for offline signatures experimentation. The images were cropped and resized to 120x120 and binarization was applied. Then DWT was applied for better feature extraction. Verification and classification was performed using naive Bayes tree, K-nearest neighbor and decision tree. The used 75% dataset in training set and 25% in testing set. For generalization of results, 10 fold cross validation was carried. Among the three classifiers, k- nearest neighbor had shown

more than 91.91% accuracy rate (without forgeries) and 88.12% accuracy rate (with forgeries). Neamah et al. [132] mined the discriminating features from offline signatures. Delphi programming language was used for experimentation. Binarization, normalization, thinning and other morphological operations were performed in preprocessing. Signature was divided into regions. COG approach was used to estimate the features for center of gravity calculation. Directed, non-weighted graph with 64 nodes was created using fuzzy logic. Optimum similarity between test and trained signature was done by dynamic classification algorithm. EER was plotted on graph to detect forgery.

Edwin et al. [168] worked on handwritten character images of special characters, Japanese katakana character, number and alphabet dataset taken from ETL-1 AIST database acquired from 1445 writers. Otsu's global threshold was used to convert the images into binary. Zhang Suen's algorithm was used to get the skeleton then connected component analysis was applied to extract curves from skeleton images of character. The curves were represented in string also the graph string structure extracted from curve feature, labeled it and stored as the knowledge. String edit distance and approximate subgraph matching was used for classification. In approximate subgraph matching method, test was performed on 1000 and 2600 sets and the training operation used 10% of each label's image. On string edit distance method, test was performed on total 8826 and 22647 sets data each data sets contains different number of images for each label. The test result on 10% training data set of number showed that the accuracy down from 82.4% into 77.7%. Accuracy was increased from 77.7% to 83.6% when using 5% data for training and string curve feature length was set as 8.

5.1.5 Other techniques

Apart from the static approaches that are discussed earlier, there are some other techniques like random forest, genetic algorithm, matching algorithms, principal component analysis, eigen vectors and quantization techniques used for the verification of the signatures. Some of the most commonly techniques deployed for the signature verification are briefly reviewed in this section. Manjunatha et al. [108] classified the signatures by using user dependent classifier. For this purpose, they experiment on six classifiers namely; Probabilistic Neural Network, SVM, Naive Bayes, Neural Network, Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA). They computed 50 global features from MCYT-75 dataset and reported 3.67% EER on skilled forgery and 0.94% EER on random forgery using 10 training samples.

Hafemann et al. [58] deployed SigNet CNN architecture for feature learning and classification. They conducted experiment on four datasets namely; GPDS, MCYT, CEDAR and the Brazilian PUC-PR dataset. Automatic features were learned from the CNN model and classified using fine-tuning technique of transfer learning. They achieved EER of 0.41 (± 0.05) on GPDS dataset, 3.40 (± 1.08) on MCYT dataset and 2.33 (± 0.88) on CEDAR dataset. Zois et al. [234] performed sparse coding on two benchmark datasets; CEDAR and MCYT75. They used 5 samples in training and branch factor of 20. An EER of 2.30% on CEDAR and 3.52% on MCYT-75 was reported. In [235], experiment was conducted on CEDAR, MCYT-75, GPDS and UTSIG datasets. They used an automated algorithm to retrieved the optimum thinning level. They extracted BRISK features using sparse representation. Furthermore, accuracy was improved by evaluating several major parameters of sparse representation. Another contribution of same author was in [233] to classified the writer independent signatures. They used CEDAR, MCYT-75, GPDS300 and D4 datasets for experimentation. Thinning and binarization was performed in pre-processing and statisti-

cal features were extracted using asymmetric pixel relation. They classified using Adaboost and achieved EER of 2.90% on CEDAR dataset, 3.50% on MCYT and 3.06% on GPDS respectively.

Roy and Maheshkar in [180] collected 36 genuine and 612 forged signatures on A4 paper. The images were binarized using OTSU's algorithm images. They calculated the boundary box and ratio between height and width. Testing was performed by breaking the images into blocks using Grid Based Approach. Contour of the signature and intersecting points of two equal half of signature was calculated by traversing the image and noted as a feature points. Centroids of two equal half of the signature were calculated using centroid based approach. Column Matching Score (CMS) and Row Matching Score (RMS) algorithm were used for verification. False Acceptance Rate (FAR) was reduced to 11-20% and False Rejection Rate (FRR) was reduced to 7-19%. Arunalatha et al. in [19] used GPDS dataset and pre-processed the acquired images using image processing techniques. Pixel Density (PD) and Centre of Gravity (CoG) were computed and data was processed using Principal Component Analysis. The threshold values of FAR and FRR between -5 to +5. ERR of proposed system was 17.08%.

Shekar and Bharathi in [197] used MUKOS database (Mangalore University Kannada Offline Signature database) and performed on a P-IV 2.99GHz Windows machine with 1GB of RAM. Total 1350 signatures collected from 30 signers in which 30 genuine signatures and 15 skilled forgeries on A4 paper. Signatures were acquired with a standard scanner with 75 dpi resolution in an 8-bit gray scale image. Morphological filter was used for binarization. Bounded regions were identified. Signature shape was subjected to PCA for the feature extraction that results in Eigen-signature. 5 samples were randomly used for training and the 40 samples for testing. Similarly, 10 and 15 signatures (randomly chosen) used as training samples and the remaining samples for testing. EER of proposed model was 14.33%. Bharadi and Kekre in [23] read pressure, spatial co-ordinates azimuth, altitude variation, Walsh coefficient of pixel distribution, code-word Histogram based on clustering technique (Vector Quantization), spatial moments of codeword, Grid and Texture features, and Successive Geometric centers of depth 2 as dynamic features of offline signature and then apply FFT, WHT and DCT and Kekre's transformation. They achieved FAR and FRR up to 2.5/8.2 and 6.5/2.96, respectively, and reported Accuracy up to 95.08%.

Kumar and Puhan in [96] proposed offline signature verification system based on partial invariant chord oriented gap (PICOG) feature. The experiment was conducted on CEDAR database of 2640 signatures samples. A majority score based approach employed to classify the signature. For training phase 7.92 threshold value selected using different sets of PICOG chords that were selected for each writer. A maximum accuracy, FAR and FRR of proposed method was 82.27%, 17.50% and 17.95%. Bisen and Mishra in [26] introduced verification with random and Skilled Forgery Detection, using Grid based Feature Extraction. The signature images were saved in Portable Network Graphics (PNG) format after scanning. Signature images were preprocessed then global, local and geometric features were extracted from resultant images. The images were converted to time series data using linear scanning and then the time series dataset is classified using shape let. The scan images were broken into grid then algorithm was used to examine peak and threshold of the each curve to the scan image for the genuine signature. The contents of signature was calculated from those cells which had 3 or more black pixels then the same process was repeated for all rows of a grid which was given those cell positions which are part of the signature image. A matrix of size $m \times n$ corresponding to the grid of size $m \times n$ was created. The matrix element was equal to 1 if the cell of same position in the grid was the part of signature, otherwise the matrix element was 0. The scan image were verified to compare with training

image whether the signatures was genuine or forged. The performance of purposed method was not explained.

Pirlo and Impedovo in [160]described work on GPDS dataset contains 24 genuine and 30 skilled forgeries gathered from 300 individuals. To detect the stable regions optical flow vectors were used to estimate the local stability of a signer, local variations were also considered for verification purposes. to estimate the low, medium, and high stability level 1800 genuine signatures (6 specimens for each one of the 300 signers) were splited into three horizontal (Initial, Central, and Final) and three vertical (Top, Middle, and Bottom) zones to train the ADT by using the optical flow vectors concerning 15 inter-class and 36 inter class variability for verification purpose. According to majority vote strategy (MVS) optimal flows between test and genuine signature was computed in testing stage. The EER of the proposed approach was 5.4% when all optical flow vectors were considered, 4.6% when medium and high stability vectors were used, and 4.0% when just high stability vectors were used. The average verification time was 0.15 s/signature. Hu and Chen [157] worked on publicly available GPDS-300 dataset and Chinese dataset. In proposed system three separate pseudo-dynamic features based on gray level i.e, local binary pattern (LBP), gray level co-occurrence matrix (GLCM) and histogram of oriented gradients (HOG) were used both in writer dependent and writer-independent mode. The Global Real AdaBoost method was used to performed classification. The classifier was trained using 10 genuine signatures and 10 random forgeries for each signer and then dissimilarity vectors for 100 writes was calculated to trained the classifier, the validation set used to determine the parameter kmax. In skilled forgery tests, all genuine signatures and skilled forgeries in the testing set were taken into account. The obtained results using CSD was 7.55% and 11.55% equal error rate.

Shah et al. in [189] present forensic analysis of signatures using offline signature sub-corpus MCYT database. The database consists of total 2250 signature in which 15 genuine and 15 forged signatures were collected from 75 individuals. Pre-processed the signature images and apply the Multi-layer perceptron and random forest classifiers for classification. The accuracy of Random forest with 5 signatures was 76.40% and the accuracy increases to 77.35% with 10 signatures and the accuracy of Multi-layer Perceptron was 66.73% with 5 signatures and the accuracy reduced to 57.98% with 10 signatures. Bharathi in [24] was conducted experiment on popular dataset CEDAR and GPDS-160. The experiments was conducted using MATLAB tool and tested on Pentium(R) dual core CPU with 3GB RAM on Windows-7. In preprocessing Otsu's method was applied on signature images then signature images were binarized using morphological operations and also performed normalization to images of size 100×200 . DCT then applied on resultant images to extract the feature set/matrix and dimension of these features reduced using bi- Linear Discriminant Analysis. MLP is trained for classification of the signature samples using the Discriminative DCT knowledge base. Two set of experiments were performed in set-1 10 genuine, 10 skilled forger signature and in set-2 15 genuine, 15 skilled forged were used for training and remaining samples for testing .The accuracy of CEDAR dataset was 92.55, FAR was 7.56, FRR was 7.65 similarly accuracy of GPDS-160 dataset was 92.41, FAR was 6.83, FRR was 7.69.

Rani and Sachdeva in [174] performed multi-modal biometric systems like speech and signature verification systems using genetic algorithm. Scale-invariant feature transform (SIFT) used for extraction of signature features while Mel Frequency Cepstral Coefficients for speech features. Image level fusion, feature level fusion, score level fusion and decision level fusion applied for authentication on various stages and reduced the extra features using genetic algorithm. Total numbers of signature and speech samples in database were 39 and the false rejected rate sample was 0 so that the accuracy of 100% was achieved. Performance evaluation of offline signature verification is tabulated in Table 3.

Table 3 Performance evaluation of offline signature verification

Reference	Features	Model/approaches	Accuracy/EER
Abuhaiba [2]	Statistical	Graph Matching	EER of 5.7%(random forgery) and 26.7%(skilled forgery)
Shanker and Rajagopalan [190]	vertical projection features	DTW and modified DTW algorithm	shown ERR of 2%(modified DTW) and ERR of 29% on DTW method.
Kennard et al. [88]	x,y and pressure	2D geometric warp algorithm	EER on English dataset was 94%–96%(blind forgeries) and 87%–91%(simple forgeries) also 74% EER on Chinese dataset.
Neela et al. [133]	X,Y trajectories, speed, angles, and their derivatives	Manhattan Distance	
Du et al. [41]	Shape context features	Hamming distance	84.15%
Roy and Maheshkar [180]	grid based feature	Column and Row Matching Score algorithm	20% FAR and 19% FRR
Edwin et al. [168]	chain code and curve features	String edit distance and approximate sub-graph matching methods	83.6% accuracy
Davda and Gonsai [36]	Local features	Distance matrices	FRR and FAR of 2.25% and 2.25%(city block distance) and FRR, FAR of 2.5% and 2.5% (square Euclidean distance)
Al-juboori [9]	Hu-invariants moments features	Euclidean distances	Statistica
Prakash and Garu [166]	relative orientation and distance geometric centroid	Max and Mean algorithm	AER of 17.33 for "MEAN" 18.26 for "MAX" fusion
Azmi et al. [21]	FCC features and global features	KNN technique using Euclidean distance	FRR of 6.67 %, 12.44 % FAR and 9.56 % AER
Malik et al. [107]	local and global features	Gaussian mixture model	EER of 20%,36% for global and local feature using ICDAR 2009 and 20%,56% for ICFHR 2010
Coetzer [32]	global features	discrete Radon transform and HMM	ERR of 12.2 %for skilled and 4.5% for simple forgeries.
Justino and Yacoubi [79]	graphometric features	HMM	

Table 3 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Kaur and Choudhary [85]	geometric features	HMM	97% accuracy
Daramola and Ibiyemi [34]	robust feature (spatial and frequency information)	HMM	99.2% accuracy
McCabe et al. [112]	Static and dynamic features	Neural Network	3.3%
Shikha and Shailja [199]	Image area, pure width and height, vertical center of the signature, horizontal center of the signature and Eigen value	MLP and SOM neural network	ERR of 18.5%
Poap et al. [163]	—	flexible neural network	85% CCR.
Sharma [196]	area, centriod coordinates, Eccentricity, Kurtosis, Skewness	Neural network	FRR=0.0366 SSIM=0.8777 MSE=66.2507 PSNR=79.8582
Wagh and Gupta [219]	60 features	BP Artifical Neural Network	
Katiyar et al. [82]	Global and local feature	ANN	3.2% FAR, 2.2% FRR and resultant errors of this system was 0.12%
Yadav and Goyal [226]	—	Neural Network	—
Khan and Dhole [89]	SURF and MSER features	ANN	—
Kumar and Dhanda-pani [94]	scale invariant and rotation invariant features	FFBP neural network	97.89 % accuracy
Dongare and Ghongade [40]	stroke, moment invariants, GLCM, color dominant and histogram	ANN	—
Padmajadevi and Aprameya [144]	angle based features	FFBP neural network	97.61 % accuracy
Kedia et al. [86]	—	ANN	average accuracy of 99.5%
Pansare [151]	Maximum horizontal and vertical histogram, Center of mass, Normalized area ,Aspect Ratio, Tri surface feature,six fold surface feature and Transition feature	FF neural network	CCR of 82.66%
Kumar et al. [97]	geometric features	Neural network	100%
Karouni et al. [81]	shape based geometric features	ANN	FAR, FRR of 1.6% and 3%
Mehra and Gangwar [113]	geometric features	Neural network	98%
Huang [69]	geometrical features	Neural network	89.24%

Table 3 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Pandey [149]	morphological features	FF Neural Network	—
Abikoye [1]	Global, mask and grid features	Neural network	100%
Sthapak et al. [206]	global, Mask and Texture features	neural network	FAR of (12.5-17.5),FRR was (10-15) and TER was (22.5- 27.5)
Anand [16]	global and statistical features	Cascaded feed-forward back-propagation networks	85-100%
Kaur [84]	Zernike Moments features	MLP based neural network and SVM	EER of 13%
Malekian et al. [106]	Aspect Ratio, Horizontal to diagonal length ratio, the slope of the line joining the center of gravity and geometric center, Vertical and horizontal variance	neural network	FRR,FAR of 4.0%, 5.3% and overall CCR was 99.17%
Jarad and Ted-mori [77]	local and global features	BP neural network algorithm	FAR, FRR, and ERR of 0.10, 0.04 and 0.415
Dash et al. [35]	pixel values in RGB format	Neural network	99.98%
Mehta et al. [114]	Sum graph and hmm features	ANN CCR of 86%-92%.	
Hatkar et al. [64]	Maximum horizontal and vertical histogram, Center of mass, Normalized area ,Aspect Ratio, Tri surface feature,six fold surface feature and Transition feature	Neural network	86.25% accuracy
Odeh and Khalil [139]	eccentricity, skewness, kurtosis and orientation features	BP Neural network	EER 21.2%
Lakshmi [99]	statistical features	Neural network model	FRR,FAR of 8%, 12% with 0.00314 mean square error
Pansare and Bhatia [151]	geometric features	BP Neural network	CCR of 85.7%-100%.
Akram et al. [6]	statistical features	FFBP neural network	100% accuracy
Sharma et al. [195]	area, centroid,eccentricity, skewness, kurtosis,orientation and tri-surface	feed-forward and Recurrent Neural Network	CCR was above 95%, FAR of 2-4% and FRR of 3-5% using supervised neural networks and CCR more than 50% FAR of 2-4% and FRR of 13-15% in unsupervised networks.

Table 3 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Puri and Verma [15]	16 radon transform based projection features	BPNN	98.03% accuracy
Mukherjee et al. [120]	—	neural network	—
Kiani et al. [92]	local features	SVM	96% FRR, FAR of 4%, 1% using Persian dataset and FRR of 19% Stellenbosch dataset
Patil and Hegadi [154]	Wavelet coefficient and detail coefficient	fast optimization Sequential Minimum Optimization (SMO) technique in SVM classifiers	FAR, FRR of 13%, 10% (linear kernel) respectively whereas 15% and 12% in non-linear case.
Pal et al. [147]	foreground/background under-sampled bitmap features, modified chain-code direction features	SVM	99.41%
Randhawa [172]	Hu's Moment Invariants and Zone Features	SVM	90% accuracy
Hu and Chen [68]	pseudo-dynamic features based on gray level: local binary pattern (LBP), gray level co-occurrence matrix (GLCM) and histogram of oriented gradients (HOG)	SVM and Global Real AdaBoost method	EER of 5.66% on GPDS and 9.94% on Chinese dataset.
Kruthi and Shet [93]	centroid, center of gravity, calculation of number of loops, horizontal and vertical profile and normalized area features	SVM	CCR was less than 7.16%
Parodi [152]	Graphometric features	SVM	FRR, FAR, EER of 7.82, 0.49 and 4.21%.
Vargas et al. [216]	Statistical texture features	SVM model	13.50% ERR.
Randhawa et al. [172]	concentric squares based geometric features, zone based slope and slope angle features	Radial Basis Function RBF-SVM	FAR of 1.25% and FRR of 1.66%.
Nguyen and Blumenstein [134]	grid based feature	SVM	AER of 13.90% and 0.02% FAR for random forgeries.
Pushpalatha et al. [167]	Polar Domain Features	Multiclass SVM	98% accuracy.

Table 3 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Pal et al. [147]	global filter based feature with G-SURF	SVM	FAR, FRR of 3.55%, 2.35% and AER of 2.95%
Hafemann et al. [60]	writer-independent features	Linear SVMs and RBF-SVMs	CCR of 14.64% on GPDS-160 dataset, EER of 10.70%, 15.05% EER with 4 signatures in the GPDS dataset and 9.83% EER with 5 signatures on the Brazilian dataset
Shah et al. [189]	—	MLP and random forest classifiers	76.40%–77.35% using Random forest and 57.98%–66.73% using Multilayer Perceptron
Bharathi [24]	Discriminative DCT features	MLP	92.55 accuracy on CEDAR dataset, FAR, FRR of 7.56, 7.65 and FAR, FRR of 6.83, 7.69 and 92.41 accuracy on GPDS-160 dataset
Armand et al. [18]	centroid, length, surface area, skew and Modified Direction Feature	Resilient Backpropagation (RBP) and Radial Basis Function (RBF) network	Accuracy of 88.0%–91.21% for RBP neural network and 88.0%–91.21% for RBF network

After the detailed work on offline signature, we elaborate the review of online signature verification system in coming section.

5.2 Online signature verification

There are number of approaches deployed in the domain of online signature verification systems just like offline signature verification systems. Some of the key contributions are explained in the coming sections.

5.2.1 Matching technique

Template matching is the process of template comparison in which test signatures matched with templates already stored in database. DTW is the most commonly used algorithm for this purpose by researchers in online signature verification system as listed below. Al-Hmouz et al. [8] conducted experiment on SCV2004 and MCYT100. They extracted dynamic features and classified them using probabilistic dynamic time wrapping (PDTW). The lowest EER of 0.002% on SCV2004 dataset and 0.012% on MCYT100 dataset reported in case 1. Durrani et al. in [43] verified the system by employing dynamic time wrapping technique. X, Y coordinates and pen up and down values of the signature was extracted from Standardized Japanese handwritten dataset. Testing performed using with and without down sampling data. They reported FAR, FRR of 27.35%, 15.18% and obtained accuracy of 79.80% respectively.

Patil and Borse in [155] worked on database having 10 users, each includes 10 genuine and 10 forged samples. Pen position data extracted from input image. Data was pre-processed and normalized and classified using DTW algorithm. The average accuracy rate of 90.4% was achieved. Fang et al. in [45] created their on database of 560 signature from 14 individuals containing 10 genuine and 30 forged samples. Initially signature images were pre-processed. T-DTW-FFT fusion algorithm modified to achieve outperformed performance using time and frequency domain features. The reported FRR of 2.86% and FAR of 1.90% with an average matching time of only 24 ms. Van et al. in [214] verified the signature using distance based approach and worked on SVC2004 database. 5 genuine signature were used for training while 15 genuine and 20 forged samples used for testing. They extracted 8 features like X, Y position, Pen status, Timestamp, Azimuth, Altitude and Pressure. They verified the signature using fast DTW technique. They achieved FRR of 0.31%, 0.51%; and FAR of 1.84%, 1.36% for Average Reference Signature and Average Permutative Difference Signature approaches.

Sharma and Sundaram in [193] used MCYT database. They extracted features like Spatial co-ordinates, Pressure, and azimuth and inclination angle. Statistical features were calculated using Gaussian Mixture Model. Derivatives were computed for wrapping the path for verification. Dynamic Time Warping matching scheme was used to enhance the verification. 3.05% EER was achieved for the common threshold. The same author in [194] used dynamic time wrapping cost matrix for verification using SVC-2004 Task 2 and MCYT-100 datasets. 5 genuine signatures were randomly selected from MCYT-100 while 20 forged and 25 skilled signatures were used for testing. 1 genuine signature per 39 users was selected from SVC-2004. Dynamic features like time sequence of attributes were extracted and DTW used for matching the test signature and reference signature and cost matrix from them was created. From DTW cost matrix cells were extracted in the wrapping path. Alignment was calculated. Normalized average distortion and displacement was computed using numerical equations. Fused score was calculated for authentication.

Maruyama and Kim in [200] conducted experiment on 1680 authentic and 1680 forged signatures from 42 subjects. All signatures were consist of kanji Characters. They processed and normalized the signatures and extracted the Inter-stroke and intra-stroke information from handwritten multi-stroke. DP matching algorithm was used for matching referenced signatures with Inter-stroke and Intra-stroke information. They achieved reduced FRR of 4.14% to 1.54%, and FAR of 4.06% to 2.01% respectively. Griechisch et al. in [54] worked on SigComp2011 database. Pre-processing was done by shifting and subtracting x, y co-ordinates from original images. Velocity value was noted as features value. The Kolmogorov-Smirnov test was applied for calculating the maximal difference between the cumulative distribution functions. KS-distance of each corresponding reference signature was calculated for training and evaluating data. They used DTW for comparison and showed FAR and FRR of 7.86%, 8.02%. The accuracy and EER about 0.50% respectively.

Impedovo and Pirlo in [72] worked on database of 50 genuine and 50 skilled forgery samples. They initially pre-processed the data and applied dynamic segmentation technique to obtain stroke-oriented descriptions. Stroke features of test samples were analyzed and estimated its stability in domain of displacement, velocity and acceleration. DTW was considered for matching points and verification performed by combining stroke verification results by simple or weighted averaging. Fischer et al. in [51] conducted experiment on MCYT online signature corpus and the SUSIG visual sub-corpus. Position, pressure, velocity, pressure derivative and acceleration features were extracted. They performed matching

using DTW and score normalization used for comparison. The reported average EER for random forgeries vary between 0.70% , 7.84% and for skilled forgeries was 2.97% and 13.79%.

Sharma and Sundaram in [192] conducted experiment on SVC 2004 and MCYT- 100. X, Y co-ordinate, pressure between consecutive points and angle-based features were extracted. Zero mean and unit variance applied to normalize features. They performed verification by comparing code vectors and verified using DTW. They achieved FAR of 1.83% and 0.78% on SVC 2004 and MCYT- 100 databases respectively. Ansari et al. in [17] proposed work using SUSIG and SVC2004 datasets. Min–max normalization and segmentation performed on samples. They extracted dynamic and shape features from each segment. To obtain a single match score for each section applied Fuzzy modeling with a mix of TS and Mamdani approaches . DTW used to find minimum distance alignment between the two samples. They achieved EER of 2.46%, 1.3% on SVC2004 and SUSIG depending on user-dependent threshold value. Mlaba et al. in [116] used SVC2004 database for experimentation. They split data in such a way 5 authentic samples used for training and 15 genuine, 20 forged signatures used for testing. They extracted features using DFT performed feature matching using Fast DTW algorithm. They reported FAR, FRR of 1.84%, 0.31% with average optimal value 0.527 for ARS and 1.36% and 0.51% average optimal value of 0.866 for APDS.

Song et al. in [204] dynamically extracted stable features. They applied optimal mother wavelet and WP reconstruction for mining spectral information from features. Modified Dynamic Time Wrapping (DTW) was employed for effective feature matching. The proposed method reported the EER of 2.89% on SVC2004 task2 database. López-garcía et al. in [103] proposed method using embedded system on MCYT-100. Signatures were pre-processed and extracted features were passed to Gaussian Mixture Model. The processed signature was align with previously stored template using DTW. They achieved EER of 2.74% respectively. Barrero et al. in [53] improved performance of online signature verification using Kinematic Theory of rapid human movements and its associated Sigma Log Normal model. DTW used for verification which works with more potential for skilled forgery analysis on BiosecurID multi-modal database. They reported 36% improvement of EER.

Rashidi et al. in [175] presented an active method using SVC2004 and SUSIG database using DCT and the parzen window classifier for matching. They extracted features like position, velocity, pressure and angle of pen using DCT. The forward feature selection algorithm used for best optimal features subsets. The EER of suggested method for SVC2004 Task1and2, Task2 and SUSIG databases were 3.61%, 2.04% and 1.49% respectively. Taherzadeh et al. in [207] presented online signature verification using C++ programming language and twenty different features. Images were scanned and preprocessed by normalization, re- sampling time and smoothing. Local and global features were extracted and feature matching performed using DTW algorithm. Best AER of proposed system achieved by combination of 5 features that was approximately 15%.

Pedrycz et al. in [7] proposed new method of probabilistic dynamic time warping (PDTW) verification techniques that consider normal DTW using SCV2004 and MCYT-100. Signature images were segmented into different segments and their possibility scores was measured with relative distance. Bayes rule used to achieve concluding results by joining these segments. The minimum improvement of PDTW parallel to the results created by the DTW method was 65%. Tahir et al. in [209] used Japanese online test set from

ICDAR2013 for experiments. Signature samples were scanned and pre-processed. They extracted x, y coordinates, pen's ups and downs positions from samples. Euclidean distance and DTW used for matching of referenced and questioned signature. The proposed system achieved accuracy of 78.14% using segmentation and 78.57% without segmentation.

5.2.2 Distance measures

Kekre and Bharadi in [87] applied Gabor filter for features extraction with time stamp and without time stamp. They used Euclidean distance method for verification. They reported EER of 90% without time stamp and EER increase up to 95% with time stamp respectively. Alizadeh et al. in [12] developed advanced method for verification based on optimal threshold selection. 62 parametric features was obtained from a digitizer plane. Conventional weighted Euclidean distance was applied for comparison purpose. They used 30 genuine and 10 forgery samples for training a considered and remaining samples for testing. They achieved FRR and FAR of 0.67%, 1.33% for 30 genuine and 10 forgery samples of training set and 2.5%, 3% for testing set.

Kaur and Kansa in [83] purposed hadamard transform based approach on SUSIG dataset. Signature images were scanned and pre-processed. Features were extracted to create hadamard matrix. Feature comparison and verification performed using Euclidean and Manhattan distance. The reported EER of proposed method was 0.05% using Manhattan 0.03% using Euclidean 0.025. Philip and Bharadi in [158] extracted features like X, Y coordinates, pressure values and Webber Local Descriptor (WLD) for online signature verification over clouds using Azure Blob storage. They passed feature vector to blob storage performed classification using K-nearest neighbor. The proposed system showed 92.50% PI (Performance Index) and 94.25% CCR.

Nisha and Deepesh in [138] developed the database of 80 signatures from 15 persons. They performed Z-score normalization on scanned data. Sequential Forward Feature Selection (SFFS) algorithm used to get 28 global features and 7 local features. They used Mahalanobis distance for verification of global features and DTW for time functions of signatures. Classification performed using KNN. The system showed 38.5% EER improved results than existing systems and 13.0% EER for skilled forgeries.

Cho and Jung in [30] introduced new method for online signature recognition using pseudo-ink. They used 105 reference images and 2625 test images data for experiments. Pen pressure value, pen tilting angle, pen theta angle were extracted and performed classification. The system showed FAR, FRR of (0.2381% , 0.2381%) and (4.7619 , 4.7619) for genuine signatures also FAR of 4.7619% for forged signature using threshold value of Pen pressure values (0.55 to 0.65).

Hafs et al. in [62] suggested method using SVC2004 and MYCT-100 databases. They used Gaussian filter, distance filter in preprocessing and extracted dynamic features using EMD approach. Similarity score was calculated by Euclidean distance for comparison. The achieved EER of proposed system was 1.83% for SVC2004 task1 and 2.23% for MYCT-100 respectively.

Nguyen et al. in [136] worked on GPDS-960 signature corpus consist of 160 sets of signature. They used Otsu's thresholding algorithm for images conversion. Data split in such a way that 12 genuine and 400 random forgeries used for testing and 12 genuine, 15 forgeries considered for testing. The Modified Direction features and the Gradient feature extracted. The Squared Mahalanobis distance and SVM used for classification. The system

produced the gradient feature produced an AER, EER of 16.52%, 16.77% for gradient feature using Mahalanobis distance on other side the AER and EER of 15.03%, 15.11% using SVM. The FAR and FRR was 16.54% and 13.51% respectively.

Yu et al. in [169] conducted experiment on MCYT-100 database. They considered 5 genuine samples for training, the remaining genuine samples and 25 forgeries used for testing. Dynamic time warping selected to co-relate sequence data. Position, pressure velocity were treated as feature vector and Mahalanobis distance applied for verification task. The system stated that MD-based method achieved higher recognition rates than ED-based method.

5.2.3 Neural networks

Iranmanesh et al. [74] used MLP neural network for classification using SIGMA database collected from 200 users. Total 4000 signature samples used for testing and training phase. Pearson correlation coefficients were selected for feature extraction. The proposed technique achieved 21.35%, 13.81% FAR, FRR and accuracy of 82.42%. Fahmy in [44] extracted Discrete Wavelet Transform(DWT) features like pen position and pen movement angle using normalization and re sampling images. They conducted experiment on SVC2004 database with 20 genuine and 20 skilled forgery signatures. 6 neural networks used for classification task. System reported 95% success rate in case of genuine signatures. Jain and Gangrade in [75] used database of 300 signature samples gather from 6 peoples. 180 samples used for training and 120 for testing purpose. Samples were pre-processed and feature extraction performed using angle, chain-Code and energy density method. The classification performed using neural network and showed accuracy of 70% to 80%.

Malallah et al. [105] proposed an approach based on ANN for using SIGMA database. A total of 200 individual signatures collected from which 10 genuine, 10 forged samples were considered for experimentation. They extracted features as time series signals of horizontal and vertical coordinates and pen pressure using PCA. Samples were normalized by up-sampling and down-sampling process. The method acquired FAR and FRR of 5.5%, 8.75% approximately. Nilchiyan and Yusof in [137] used SVC 2004 database from which 15 genuine, 15 forged signatures used for training. The statistical features extracted using wavelet transformation. B-spline function used for the better quality and cost reduction. Then Multi perceptron neural network used to classify the signatures and the overall 3.5% EER was achieved.

Xu et al. in [225] proposed new method using BP neural network for signature identification. Total 400 signatures collected from 8 volunteers. Total of 49 features were calculated and used as feature set. Final acquired recognition accuracy of proposed system was 95% approximately. Iranmanesh et al. in [74] introduces Pearson correlation coefficients method for feature extraction of online signatures. They selected 10 genuine, 5 skilled and non skilled forgeries for training and testing phase from SIGMA database. 9 different features were extracted and verification performed using Multilayer perceptron. The achieved FAR, FRR and accuracy of proposed approach were 21.25%, 13.81% and 82.42%. Chadha et al. in [29] worked on database of 700 samples by taking 10 samples from 70 people. Initially rotation-translation-scaling was applied for images processing. Feature were extracted by DCT and the signature identification performed using RBFN. The system achieved a recognition rate of 80% for approximately 200 samples.

5.2.4 Hidden Markov model

Nanni et al. [125] presented new verification system based on combination of local, global and regional matchers using MCYT signature database. The DTW, HMM and a Linear Programming Descriptor have taken into account as matching techniques. Using Bio-Hashing and Bio-Convolution approaches with a template protection scheme the EER was 4.51% which is very low. On the other hand proposed matching scheme give more effective result of 3%.

5.2.5 Support vector machine

Durrani et al. in [42] conducted experiment on ICDAR 2013 signature images using SVM classifier. They considered 396 forged samples for training whereas 42 genuine and 36 forged signatures per author provided for testing. The obtained threshold value was 1055. Dynamic feature set were extracted and stored in database. The system achieved EER of 19.98% and accuracy was more than 75%. Khoh et al. in [90] organized study on SVC2004 Task 1 and Task 2 datasets. The images were scanned and pre-processed. DFT, DTW used to extract feature set from each signature signal. They used Euclidean distance and EED for matching the feature of test and reference signature. Verification done using KNN and SVM. They achieved EER of 6.55%, 1.29% for skilled signatures and random forgeries in Task1. While for Task2 dataset EER of 7.17%, 1.15% for skilled and random signature was achieved.

Griechisch et al. in [55] developed new technique for verification by comparing AccSigDb2011, GyroSigDb2012 databases comprised of local acceleration and angle information. They used Legendre approximation for feature calculation and classification performed using SVM. The average accuracy is around 85%-89% in AccSigDb2011 whereas the GyroSigDb2012 accuracy was less. While using multiclass classification %, the accuracy of GyroSigDb2012 was 50% while 35-40% on the AccSigDb2011 if the order of the Legendre polynomials order was less than 20. Rosso et al. in [179] conducted work on MCYT data base used for quasi-offline signatures. The information of signature coordinates considered as quasi-offline. The images were scanned and performed interpolation to extract time series of both horizontal and vertical writing processes. Six-dimensional vector were extracted and built histogram for each coordinate. Classification performed using Support vector machine. The EER of proposed method using 5, 10 training samples were 0.19% , 0.17%, respectively.

Manjunatha et al. in [109] performed experiment on MCYT-100 and MCYT- 330. 100 global features were considered. 6 different statistical classifiers as Naive Bayesian, Nearest Neighbor , Support Vector Machine, Principal Component Analysis, Linear Discriminant Analysis and Probabilistic Neural Network used for classification. They used 5 and 20 genuine signatures per writer for training set. They reported FRR, FAR and EER of 3.83%, 0% and 1.92% respectively. Li et al. in [101] acquired 2,940 electronic signatures of 42 participants on smart phone. 57 features were extracted using Python and 57-dimensional feature vector having 50 positive samples feature vectors and 20 negative samples feature vectors was created. SVM, Logistic Regression, AdaBoost and Random Forest were used for the classification. The system achieved Error rate of 2.375% using AdaBoost. Fayyaz et al. in [47] conducted experiment on SVC2004 database. Images were scanned, normalized and dimensions reduction performed using PCA. Features were extracted using unsupervised

learning. SVM with RBF and KNN classifiers were used for verification. They tested experiment on 1600 signatures. The proposed model achieved EER of 2.15% and 99.11% with KNN for 50% training set.

5.2.6 Structural approaches

In the structural approaches trees, graphs and strings are used to exhibit patterns of the samples. The symbolic representation will be computed with patterns restored in the network.

Wang et al. in [223] proposed a method using SUSIG database. Initially signatures were represented by a series of graphs then Graph matching techniques presented to calculate Edit distance between graphs to measure likeness among graphs. Classification performed by user-dependent classifiers. The achieved average error rate of proposed and Fusion with DTW were 5.80%, 2.46%. Shah et al. in [188] performed comparative analysis and employed three classifiers like decision tree (J48), naive Bayes (NB tree) and KNN for verification. They used ATVS-SSig datasets which contains 350 signatures. The DWT used for features extraction. The method achieved 99.90% accuracy with decision tree, 99.82% with naive Bayes and 98.11% with distance based k- nearest neighbor classifiers.

Pirlo et al. in [159] organized study on SUSIG database from which 10 genuine and 5 skilled forgeries exploit for training and testing. Linear normalization Algorithm and linear interpolation algorithm were used for preprocessing. Displacement, velocity, acceleration and pressure function parameters extracted from samples. DTW used to perform prototype selection and to detect the wrapping function. Then defined the stability model of the signer and performed verification using decision tree. Standard deviation method used to calculate signature difference and Probabilistic acceptance model practised for performance enhancement. They achieved FRR and FAR of 2.15% and 2.10%.

5.2.7 Other techniques

Alpar and Krejcar in [14] used spectrogram analysis for online signature verification. An interface was written in Matlab that consists of signatures silhouette with 600×355 JPG file. Hamming distance was used for the calculation of distance that is converted into discrete signals. Short-time Fourier transforms (STFF) was used for the creation of 2D RGB images. Quantization was used to find grid histograms of signature sample. They used Fuzzy surface for classification. They reported EER of 4.82% on threshold value 90 by linear interpolation. Tahir and Akram in [208] worked on online Japanese dataset from ICDAR2013 collected by HP Elite Book 2730p tablet PC and self-made collection software built with Microsoft INK SDK. Sampling rate was 200Hz. 11 signers participated for training set and 20 signers participated for evaluation set. Resampling and normalization performed on captured signatures. Fourier transform based features, Wavelet transform based features and global features were extracted and combined for performance evaluation. The training set consists of 12 genuine reference signature and approximately 64 questioned signatures. The method achieved FRR of 25.54% respectively. Nambiar et al. in [123] introduced locking system based on online signatures. Signature images were captured using touch based mobile. they extracted features like number of pixel points, signature length, speed and velocity. Features were saved in the database. For registration 5 signature templates were taken from user for login into the system.

Daramola and Ibiyemi in [33] collected signature samples using Graphics tablet as capturing device. The Samples were pre-processed then extracted seven features. Authentication depend on the average of all the distances accomplished from the cross-alignment of the features. The proposed system presented error of 0.25% in rejection of genuine signatures and 0.5% for skilled forgeries. Al-mayyan in [10] worked on database of 2160 samples collected from 108 individuals. 31 global features were extracted from which minimal set of nine features were taken using PCA and Rough sets. Classification performed using Naïve Bayes and Rough set classifiers. Rough set classifier achieved 100% CCR. Yang and Liu in [228] used SVC2004 database for verification based on Gaussian mixed models. Signature comprised of 7-dimensional information were captured and pre-processed by eliminating duplicate points, removing forgery and Gaussian filter. Wavelet packet analysis used to extract local features and combination of global features used as feature set. Splitting-EM algorithm based on Bayesian Ying-Yang learning system to trained the GMM to achieve excellent no. of Gaussian components. Different online verification systems along with results are reported in Table 4.

6 Trends and challenges of signature verification system

Handwritten signatures can be openly challenged at any time which is not possible in other biometric systems. Thus, despite extensive research signature verification is still remains open to the research community due to the variety of challenges it offers. In this section, we highlight some of the notable challenges of handwritten signature verification systems that pave the path for future research in this domain.

Starting from the image acquisition phase, one of the major issue is the lost of dynamic information in offline signature generation process. The offline signatures are captured by scanning the document that contain signature information. During scanning, the dynamic information like pen position, speed and velocity of writing is often lost that leads to the lowest identification rate and makes the problem challenging. However this challenge is overcome in online signature verification system, where the signature information is captured through online devices like digitized pens and tablets, PDA, graphic tablets, and even through online gloves. The dynamic information is provided in online systems but the availability of online devices is still an important factor. Furthermore, the online devices should ensure the certain standard for providing best accuracy, authenticity and system legitimacy. These standards vary from country to country like in Europe, electronic IDentification, Authentication and trust Services (eIDAS) standard is used.

The signature verification system inherits an important challenge from writer recognition system i.e. large intra-class variations. Like handwriting, signature of a person may vary from certain factors like age, physical or mental states, mood and situation etc. These factors makes the system more challenging and difficult because the intra-class variations allows the system to verify the skilled forged signature as the original genuine signature to a very great extent. Intra-class variations may produce in offline or online signature verification system. In offline signature verification system, signatures were acquired from different equipment thus the resolution of training and testing samples are not same and leads to the intra-class variation. On contrary, in online signature verification, sometimes the online devices like tablets, mobile devices, PDAs and touch screens are characterized

Table 4 Performance evaluation of online signature verification

Reference	Features	Model/approaches	Accuracy/EER
Alizadeh et al. [12]	62 parametric features	Conventional weighted Euclidean distance for comparison	FRR,FAR of 0.67%, 1.33% for training set and 2.5%, 3% for the testing set
Durrani et al. [43]	X,Y co-ordinate and pressure	Matrix based approach	FAR, FRR of 27.35, 15.18 and 79.80% accuracy.
Kaur and Kansa [83]	Hadamard transform	Manhattan Euclidean distance	EER of 0.05
Patil and Borse [155]	pen movement and pen position angles	Dynamic Time Wrapping	90.4% accuracy.
Philip and Bharadi [158]	X, Y coordinates, pressure values and Webber Local Descriptor	K-nearest neighbor	94.25% CCR.
Cho and Jung [30]	pen pressure value, pen tilting angle, and pen theta angle	K-NN classifier	the FAR, FRR of (0.2381-0.2381) and (4.7619-4.7619) for genuine signatures and FAR of 4.7619 for forged.
Fang et al. [45]	time and frequency domain features	T-DTW-FFT fusion algorithm	FRR,FAR of 2.86%,1.90%
Van et al. [214]	x position, y position, pen status, timestamp, azimuth, altitude and pressure	fast DTW	FRR of 0.3-0.51 and FAR of 1.84-1.36
Sharma and Sundaram [193]	spatial co-ordinates, pressure, and azimuth, inclination angle and Statistical features	DTW matching scheme	EER of 3.05%
Sharma and Sundaram [192]	Dynamic features like x and y co-ordinates, pressure, azimuth and inclination angle	DTW	
Maruyama and Kim [200]	Inter-stroke information and intra-stroke information	DP matching algorithm	FRR reduce from 4.14% to 1.54% and FAR from 4.06% to 2.01%.
Nisha and Deepesh [138]	28 global features and 7 local features	KNN classifier	overall EER of 38.5% and 13.0% for skilled forgeries
Griechisch et al. [54]	x, y, p and v velocity vale	KS-distance distance	FAR, FRR of 7.86, 8.02 and EER of 0.51.0%.
Impedovo and Pirlo in [72]	stroke-oriented features	Dynamic Time Warping	
Tahir and Akram [208]	Fourier transform based features, Wavelet transform based features and global features	matching(reference and questioned signature)	FRR of 25.54%.

Table 4 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Fischer et al. [51]	Position, pressure, velocity, pressure derivative and acceleration	Dynamic Time Warping	EER between 0.70% and 7.84% for random forgeries and between 2.97% and 13.79% for skilled forgeries
Sharma and Sundaram [192]	X co-ordinate, y co-ordinate and pressure between consecutive points, and angle-based features	Dynamic Time Warping	FAR of 1.83%, 0.78% on SVC 2004 and MCYT-100 dataset
Tahir et al. [209]	x, y coordinates of the pen and pen's ups and downs positions	Euclidean distance and DTW	78.14% accuracy using segmentation and 78.57% without segmentation
Ansari et al. [17]	Shape and dynamic features	DTW technique	EER of 2.46% using SVC2004 and 1.3% using SUSIG
Lopez-Garca et al. [103]	x, y position and pressure	GMM	EER of 2.74%
Mlaba et al. [116]	x, y position, pen status, timestamp, azimuth, altitude and pressure	FastDTW algorithm	FAR, FRR of 1.36, 0.51.
Song et al. [204]	stable features	Modified DTW	2.89% EER
Barrero et al. [53]	Sigma LogNormal features	DTW-SL approach	—
Rashidi et al. [175]	position, velocity, pressure and angle of pen	parzen window classifier	ERR of 3.61%, 2.04% for SVC2004 Task1/2 and 1.49% for SUSIG databases.
Taherzadeh et al. [207]	Local and global features	DTW algorithm	AER of 15%
Pedrycz et al. [7]	x position, y position, pressure p, azimuth z, and altitude of the pen	DTW and PDTW	65%.
Yang et al. [228]	local, global features	Gaussian Mixture Model	95.8 accuracy%.
Fahmy [44]	like pen position and pen movement angle	neural networks	95% accuracy
Li et al. [101]	57 features (current number of writing times, current time, coordinates x, y of position, pressure, contact area, status of current position etc)	SVM, Logistic Regression, AdaBoost and Random Forest	EER of 2.375%
Jain and Gangrade [75]	Energy, Angle and Directional Gradient Feature	Neural network	70-80% accuracy
Durrani et al. [42]	Dynamic feature	SVM	ERR of 19.98% and accuracy 75%.

Table 4 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Malallah et al. [105]	Time series signals of horizontal and vertical coordinates and pen pressure	ANN	FAR of 5.5% and FRR of 8.75%
Manjunatha et al. [109]	100 global features	Naive Bayesian (NB), Nearest Neighbor (NN), SVM, PCA, Linear Discriminant Analysis (LDA) and Probabilistic Neural Network	FRR, FAR of 3.83, 0 and EER of 1.92
Khoh et al. [90]	x and y coordinates, time stamp, and pen-up status of a signature	KNN and SVM	7.17% EER for skilled signatures and 1.15% EER for random signature
Nilchiyan and Yusof [137]	statistical features	Multi perceptron neural network	3.5% EER
Griechisch et al. [55]	—	SVM	Average accuracy is around 85%-89% in AccSigDb2011 using multiclass classification % accuracy of GyroSigDb2012 was 50% while 35-40% on the AccSigDb2011.
Xu et al. [225]	49 features	BP neural network	95%
Nanni et al. [125]	x,y position, pressure, altitude, azimuth,	Dynamic Time Warping, Hidden Markov Model based approach and a Linear Programming Descriptor classifier	EER of 4.51%
Wang et al. [223]	—	user-dependent classifiers(graph matching)	AER on single System and Fusion with DTW were 5.80%, 2.46%.
Shah et al. [188]	—	decision tree (J48), naïve Bayes (NB tree) and KNN	achieved 99.90% accuracy with decision tree (J48), 99.82% with naïve Bayes (NB tree) and 98.11% with distance based k- nearest neighbor
Pirlo et al. [159]	Displacement, velocity, acceleration and pressure function parameters	decision tree classifier	FRR of 2.15% and FAR of 2.10%.
Alpar and Krejcar [14]	frequency features	Fuzzy surface	82% EER

Table 4 (continued)

Reference	Features	Model/approaches	Accuracy/EER
Daramola and Ibiyemi [33]	Δx , Δy , Δp , $\Delta y/\Delta x$, $\Delta p/\Delta x$, $\Delta p/\Delta y$ and v	average distance threshold technique	Error of 0.25% in rejection of genuine signatures and 0.5% for skilled forgeries.
Hafs et al. [62]	dynamic features	Euclidean distance	EER of 1.83,2.23% for the SVC2004 task1 and MYCT-100 dataset
Kekre and Bharadi [87]	time stamp and without time stamp features	Euclidian distance method	EER of 90% without yime stamp and 95% with time stamp
Iranmanesh et al. [74]	Pearson Correlation Features	Multilayer perceptron (MLP) NN	FAR of 21.25%, FRR of 13.81% and accuracy of 82.42%.
Nguyen et al. [136]	Modified Direction Feature and the gradient feature	squared Mahalanobis distance classifier and SVM	AER of 16.52% and an EER of 16.77% and the AER of SVM was 15.03% and EER was 15.11% using gradient feature . The FAR and FRR respectively are 16.54% and 13.51%.
Yu et al. [169]	position, pressure and velocity	Mahalanobis distance	—
Al-mayyan [10]	31 global features	Naïve Bayes and Rough set classifiers	100% accuracy with Rough set
Fayyaz et al. [47]	X, Y coordinates, time and pen up/down, azimuth, altitude and pressure	SVM with RBF (radial basis function), KNN and PCA	99.11 (with KNN)
Chadha et al. [29]	X, Y coordinates, pressure etc.	RBFN	80% accuracy
Iranmanesh et al. [74]	x,y coefficient, pressure	MLP neural network	FAR, FRR of 21.35%, 13.81% and accuracy of 82.42%.

by a small input area and poor ergonomics. These factors affect the user interaction and a large intra-class variation is produced.

Moving towards the discriminative feature extraction and classification phase, researcher often a variety of challenges. Among them one of the challenge is the extraction of discriminative visual features. Literature divulges that a lot of work is presented using handcrafted features and traditional classifiers. However, deep learning techniques were not explored at the higher level. A reason might be the availability of the large amount of dataset used for enrollment/training process, as CNN model requires a numerous amount of data samples in training. In most of the real world applications, only a small amount of data samples are used. Therefore, the discriminative visual features are not extracted using CNN models. Apart from the CNN models, large dataset is also required for real applications especially

in forensic analysis and banks documents. Generally the classification is performed on the limited data samples thus new user is not verified exactly. If there is a large number of samples for the training of the system, a classifier needs to perform well for a new user, for whom only a small set of samples are available.

The major objective of an automated signature verification system is to discriminate the genuine signature from the pre-stored signatures to reduce the forgery. Mostly, the data samples used for training the system contains only the information of the genuine signatures. These systems verify the genuine signatures but reject to provide the details of forgeries. This leads to the challenging task, as classifier is not trained on the forged data. There should be the dataset for automatic signature verification system that contains the complete information of genuine and forged signatures rather than the partial information of genuine signature.

Another important challenge inherited from writer recognition system is the multiscript signature verification systems. Different scripts and design for signatures have unique appearance, therefore it's very hard to build up a complete system that classify every script and style because feature selection depends on script and style of signatures. There are two basic fundamentals of multiscript signature verification systems. Firstly, the signatures of multiple languages are verified by the same automatic system. Secondly, to validate the hypothesis that signer across different scripts are same.

7 Conclusion

In this paper we disclosed a brief survey on signature verification to climax the state-of-the-art to researchers of the signature verification and pattern recognition field. It has been examined that line of work has already done in the field of signature verification but there are many challenges in this area, due to which still it is most hot issue in document analysis, computer vision and image processing. The survey covers techniques of signature verification along with offline and online domain.

The highlighted challenges of Section 6 tends a reader towards the key future directions. Their is a need to develop the offline database or system that will kept the dynamic information. This will achieved from the availability of acquisition devices and the development of standards. In online signature verification, due to the large variety of electronic acquisition devices, another relevant future direction concerns with the problem of Interoperability. The Interoperability will be achieved through enhanced preprocessing and robust features selection techniques. The signature information from different electronic devices will be normalized in a specific range and then robust features will be extracted. This will ensure inter operatability to the great extent. Furthermore, their is a need to develop the standards for electronic devices like in Europe, electronic Identification, Authentication and trust Services (eIDAS) standard is used. Unfortunately in Eastern countries, electronic devices and their standard are not established at higher rank. Another important direction is the development of large dataset that contains the information of genuine as well as forged signatures information. The available dataset in the literature only provides the information of genuine signatures. In real world applications like in forensic analysis, fraud detection, cheque processing etc. forged information is required to detect the forged person. There should be the dataset for automatic signature verification system that contains the complete information of genuine and forged signatures rather than the partial information of genuine signature. Their is a large room for the development of multi-modal biometric based system. Their should be the system that can identified the person using signature as well as from any other

biometric modality. This need arises in real world domains where a person injured or suffer from accident. In this accidental situations, signature is not sufficient to verify the person.

In future work, we aim to explore deep learning and transfer learning techniques. The deep learning and transfer learning techniques proved itself as a best feature extractor in the handwriting recognition, writer recognition, optical character recognition and speech recognition etc. Lastly, advances are required in offline and online multi-script signatures, which remains an open issue for the researchers of the domain.

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